University of Reading

Department of Computer Science

The Psychological Narrative Engine (PNE)

Jerome Bawa

Supervisor: Pat Parslow

A report submitted in partial fulfilment of the requirements of the
University of Reading for the degree of
Bachelor of Science in *Computer Science*

March 30, 2026

Declaration

I, Jerome Bawa, of the Department of Computer Science, University of Reading, confirm that this is my own work and figures, tables, equations, code snippets, artworks, and illustrations in this report are original and have not been taken from any other person's work, except where the works of others have been explicitly acknowledged, quoted, and referenced. I understand that if failing to do so will be considered a case of plagiarism. Plagiarism is a form of academic misconduct and will be penalised accordingly.

I give consent to a copy of my report being shared with future students as an exemplar.

I give consent for my work to be made available more widely to members of UoR and public with interest in teaching, learning and research.

Jerome Bawa
March 30, 2026

Abstract

Non-player character (NPC) dialogue in video games has historically been limited by branching conversation trees, a system that prioritises authored consistency at the cost of depth, immersion and agency. This project presents the Psychological Narrative Engine (PNE), a dialogue system that replaces scripted branching with a Belief-Desire-Intention (BDI) processing pipeline, enabling NPC responses to be emergent via models of internal cognition and social identity rather than from pre-authored text.

The system bases NPC cognition on a three-layer personality model and thought library, then outsources the text generation to a local large language model (LLM) via Ollama. The remainder of the system includes a gamified skill check, a judgement-score finite state machine and game engine agnostic REST and WebSocket API.

The central question investigated is whether psychological realism and narrative coherence in NPC interaction can be achieved simultaneously with player agency. Evaluation is conducted through structured conversation log analysis and internal state inspection across representative test scenarios. The system targets players running mid-to-high end consumer hardware, a constraint imposed by the memory requirements of local LLM models at scale.

Contents

Chapter 1: Introduction

1.1 Introduction

The dialogue tree has defined NPC interaction in commercial video games for three decades. From video games such as *Fallout 4* (Bethesda Game Studios, 2015) to *The Walking Dead* (Telltale Games, 2012), the dominant model has remained the same. A narrative designer's task is to author every exchange. The player selects from a fixed set of options, and the system traverses a predetermined graph. This approach produces narratively consistent and commercially polished experiences. However, it also features a limitation — one that scales poorly and fails to address the psychological nature of the characters it represents.

For each new conversational branch, demands an increased investment in writing, quality assurance and voice acting. These resources compound quickly. More critically, the dialogue tree is psychologically hollow. NPCs in these systems do not hold beliefs or form intentions; they evaluate conditions and return pre-written strings. They do not reason about what a player has said. The apparent depth of a well-written branching tree is purely a product of human authorship, not of any computational representations of cognition.

This project begins with *Disco Elysium* (ZA/UM, 2019). That game produced a quality of emergent narrative experience that the dialogue-tree model cannot replicate; but it did so by directing the player's experience of the narrative emergently. The player character's internalised skills manifest as competing perspectives, intrusive thoughts and divergent interpretations of the same event. The emergent experience in *Disco Elysium* is not the product of NPC behaviour; it is the representation of how reality is not objectively experienced but subjectively understood.

This raises a distinct and underexplored question: is it possible to locate that same quality of psychological emergence on the *NPC* side? Can an NPC hold an internal model of the world, form goals in response to what a player says, and select a communicative behaviour that expresses that internal state. Is it possible to be achieved without sacrificing narrative coherence or reducing player agency in ways previous video games have done?

This dissertation argues that such a system is now feasible and documents the Psychological Narrative Engine (PNE) as a prototype demonstration of that feasibility. The PNE models each NPC through a three-layer personality structure; cognitive, social, and world knowledge. It processes each player turn through a staged BDI pipeline: belief updating, desire formation, social intention selection, and outcome application. The LLM's role is narrowly scoped to surface text generation — producing the spoken words of the NPC, controlled by a closed set of pre-authored behavioural intentions. The separation of what the NPC thinks from how the NPC speaks is the central architectural claim of this work.

The PNE is implemented as an API server with a REST and WebSocket API, which can connect to game engines like Unity, Godot and Unreal Engine without locking the dialogue to any game engine. A two-dice skill-check system, like *Disco Elysium*, resolves the success of player dialogue turn choices, with the probability of success calculated from player skills, NPC personality, and player/NPC relationship state and difficulty settings. A decision-score-based finite state machine (FSM) aggregates dialogue outcomes, across turns, to a stable 0-100 decision score that is mapped onto narrative outcomes, independent of the LLM output.

1.2 Motivation

1.2.1 The Research Problem

The core investigation of the project is the balance between coherence, realism, and player agency in NPC dialogue systems. These three properties have previously been in partial conflict. A system that prioritises coherence will rely on tight authorial control, limiting emergence. Systems that allow emergence tend to sacrifice coherence, and systems that maximise player agency often do so by reducing NPC behaviour to reactive outputs with little to no internal modelling.

The research question is this: *how far can a BDI cognitive architecture, implemented over a formalised NPC personality model and a constrained large language model (LLM) text generation, create NPC dialogue that is psychologically realistic, narratively consistent and responsive to player input?*
This question has become possible to explore for a specific reason: locally deployable large language models through services such as Ollama. Dialogue text of sufficient quality for game integration can now be generated on consumer hardware during real-time play. This opens a space that was previously inaccessible — reasoning and text generation can be decoupled. The reasoning layer operating deterministically under developer control. The generation layer producing natural language based on the results of that reasoning. The PNE is designed to occupy this space.

1.2.2 Commercial and Economic Context

This project's commercial success would not be constrained by the size or funding of the developer, but by the hardware capabilities of the player. The PNE can be used by a wide range of developers from the smallest indie studios to medium sized teams. It's a middleware service with no cloud dependency and no cost-per-call. The limitation is on the client as the system assumes a player machine capable of running local LLM instances, in addition to the game process.

The technology that defines this constraint is Ollama. Models capable of generating dialogue of a quality to be integrated into a game (such as Llama 3 or Mistral in the 7B-13B range) require at least 4-6 GB of VRAM for inference at acceptable rates. It is not intended to run on integrated graphics hardware, low-end laptops or mobile devices. Cloud-hosted inference is deliberately excluded from the architecture, motivated by the latency requirements of real-time dialogue streaming and the undesirability of a persistent network dependency for an offline-capable game feature.

Importantly, this hardware restriction does not limit the PNE to large productions. It enables developers who would not have the resources to author deep NPC dialogue. A game like *Stardew Valley (ConcernedApe, 2016)* was created by a single individual, with named characters, schedules, relationship-building and recurring dialogue. This is the type of project where dynamic, psychologically realistic NPC dialogue would enhance replayability and emergent gameplay the most.

Authoring all the villagers as dynamically adaptive dialogue systems is too much of an authorial burden for a single author; the PNE allows an author to input personality configurations and scenario JSONs, so the system generates dialogue. This makes the PNE not a tool for a studio that has a dialogue designer, but a system that allows a studio that might not have enough dialogue designers to have psychologically realistic NPC dialogue - assuming that the people playing the game have the hardware to play it.

1.3 Outline

The project begins by establishing the limitations of existing NPC dialogue systems. A literature review then surveys multiple bodies of work informing the design: cognitive psychology models of belief and inference, social psychology frameworks covering persuasion and impression management, BDI agent architectures, and prior work in interactive narrative.

The design and implementation of the Psychological Narrative Engine is then documented in full, covering the seven-stage BDI pipeline, the *CognitiveThoughtMatcher*, the skill-check system, the judgement-score finite state machine, and the REST and WebSocket API. Chapter 5 analyses the system design; Chapter 6 presents conversation log results across three NPC profiles. Chapter 7 evaluates these against the system's central claims. The project concludes with a reflection on its achievements, limitations, and directions for future development.

Chapter 2: Social, Legal, and Ethical Considerations

2.1 Overview

The Psychological Narrative Engine operates at the intersection of artificial intelligence, cognitive modelling, and interactive entertainment. As with any system of computational modelling of psychology for consumer applications, the development and use of the PNE has social, legal and ethical consequences.

Ultimately, the social implications of the PNE are judged as net positive, especially for those smaller developers who have until now been unable to take advantage of deep NPC narrative.

2.2 Social Considerations

2.2.1 Impact on Smaller Studios and Independent Developers

The primary social benefit of the PNE is democratisation of narrative depth. Authoring psychologically rich NPC dialogue at scale has historically required dedicated writing teams, and other resources that smaller studios cannot sustain. The result is a structural disparity in narrative quality between large and small productions. This is little to do with creative ambition and everything to do with production resources.

The PNE disrupts this disparity by shifting what must be authored. A solo developer or small team can define NPC personality configurations and scenario graphs without writing individual dialogue lines. The system generates surface dialogue from that structure, producing adaptive, contextually responsive NPC behaviour at a fraction of the resource investment. For games with social simulation at their core, this represents a meaningful capability expansion. The class of game exemplified by Stardew Valley (ConcernedApe, 2016) stands to benefit most directly: titles where NPC depth is central to the experience, but the production scale is small.

2.2.2 Impact on Narrative Designers and Industry Labour

The question of whether AI-assisted dialogue generation displaces narrative designers depends substantially on the scale of the studio in question.

For smaller studios that currently cannot employ narrative designers at all, the PNE does not displace anyone — it fills a gap that was previously unfilled. It enables narrative experiences that would not otherwise exist, and in doing so may create demand for narrative design consultation on configuration and scenario authoring that did not previously exist at that scale.

For mid-scale studios with small writing teams, the impact is a likely shift in workload rather than a reduction in headcount. Narrative designers working with the PNE would move from writing individual dialogue lines to authoring personality models, thought-pattern templates, and scenario JSONs — a more systems-oriented form of narrative design. This is a change in the nature of the work, the skill set required evolves rather than disappears. Whether the headcount evolves with it is less certain, however. A studio may conclude that a single systems-thinking designer is sufficient instead of two or more dialogue writers.

For large studios such as *Ubisoft, Bethesda Game Studios, and Rockstar Games* — the impact of the PNE is likely minimal. These studios do not prioritise emergent NPC dialogue as a design goal. Their flagship productions (*Assassin's Creed*, *The Elder Scrolls*, *Grand Theft Auto*) are built around authored story arcs, scripted set-pieces, and fixed character trajectories that require precisely the kind of controlled narrative authorship that large writing teams provide. The PNE is not designed for this mode of storytelling and would offer these studios limited utility in their current production pipelines. If the PNE is to be beneficial, it is most likely to be as an additional system rather than a flagship part of the product, therefore designers are not at risk from a displacement due to the PNE.

The industry trend towards AI-powered content generation is a cause for concern for creative labour of all kinds, but the PNE is a structured intervention and not a general creative AI.

2.3 Legal Considerations

2.3.1 Software Licensing and Open-Source Compliance

The PNE is built on a stack of open-source components: FastAPI (MIT licence) and the Ollama runtime (MIT licence). The model used in the prototype implementation is Qwen2.5:3b (Qwen Team, 2024) and distributed under the Apache 2.0 licence. This is among the most permissive open-source licences available — it permits use, modification, and commercial distribution without royalty obligations, provided attribution is retained. This makes Qwen2.5:3b a legally straightforward choice for both research and any subsequent commercial integration built on the PNE.

Qwen2.5:3b's 3-billion parameter scale is notable from a deployment standpoint: at 4-bit quantisation the model requires approximately 2 GB of VRAM, a significantly lower threshold than the 7B–13B models more commonly referenced in the literature. This broadens the viable player hardware base compared to larger models, though dialogue quality trade-offs relative to more capable models are acknowledged and discussed in Chapter 5.

No proprietary third-party APIs are called at runtime. The system generates no logs containing personal data, makes no external network requests during normal operation, and does not depend on any service whose terms of use could change post-deployment. Developers substituting a different Ollama-compatible model in a production integration carry responsibility for verifying the licence terms of their chosen model independently.

2.3.2 Intellectual Property

The PNE does not reproduce copyrighted text, dialogue, or narrative content. NPC dialogue is generated at runtime from LLM inference constrained by developer-authored configurations; no pre-existing authored content is replicated. The thought-pattern template library is original work. Scenarios and personality configurations authored by developers who integrate the PNE remain the intellectual property of those developers.

The use of *Disco Elysium* as a design reference for the skill-check mechanic is analytical and transformative — the mechanic is independently implemented and does not incorporate any code or content from ZA/UM's work.

2.3.3 Data Governance

There is no personal data that is collected, stored or transmitted. No player identification is performed; session state is held in memory for the duration of a conversation and discarded thereafter unless explicitly persisted by the game engine accessing the API. No questionnaires or user studies were conducted in the development of this project, and no ethical approval for data collection was required or sought.

2.4 Ethical Considerations

2.4.1 Representation of Psychological Constructs

The PNE applies theory drawn from cognitive psychology including those about belief, desire, intention and emotional state. These constructs are used symbolically within a game context and are not intended to constitute clinical representations of human psychology. NPC personality parameters are design tools, not accurate psychological assessments. The system does not present itself as a clinical model of real human cognition.

This distinction is important to maintain clearly in any production context. Game NPCs modelled through the PNE should be understood as fictional characters whose behaviour is shaped by design parameters, not as simulations of real psychological conditions. Developers integrating the system carry responsibility for ensuring that NPC characterisation does not reproduce harmful stereotypes or present psychologically distressing content without appropriate content warnings.

2.4.2 Transparency of AI-Generated Content

Dialogue generated by the PNE is produced by a LLM at runtime. Players interacting with PNE-driven NPCs are engaging with AI-generated text, constrained and shaped by developers, but not authored line-by-line by hand. The ethical question of whether players should be informed that NPC dialogue is AI-generated is a live one within the games industry and the broader AI content space. This project takes no position on such obligations, which is a matter for the integrating developer. It is noted, however, that transparency about AI-generated content is viewed as the best practice and is recommended.

2.5 Summary

The social, legal, and ethical profile of the PNE is primarily mild. Its primary social effect is to extend narrative design capability to developers who currently lack it, with a secondary effect of shifting the workload of narrative designers at mid-scale studios. Legal compliance is straightforward given the open-source stack and absence of data collection. The core ethical obligations: responsible characterisation and transparency about AI-generated content become responsibility of developers rather than with the engine itself.

Chapter 3: Literature Review

3.1 Overview

The PNE synthesises four domains of research: cognitive psychology, social psychology, BDI agent architectures and interactive narrative. Each field provides a conceptual component of the system design, and each has its own limitations that need to be considered. This chapter examines the literature in these fields, assesses its applicability to computational modelling of NPCs and concludes with the place of the PNE in this context.

3.2 Cognitive Psychology Foundations

3.2.1 Constructive Cognition: Neisser (1967)

Neisser (1967) established the theoretical basis for treating perception and memory as constructive rather than reproductive processes. Where earlier models had characterised mental activity as the passive accumulation and retrieval of stored impressions, Neisser argued that every cognitive act involves an active reconstruction. Neisser's introduction of schema theory, the claim that perception is mediated by pre-existing interpretive frameworks built from experience was foundational to contemporary cognitive science. Two people can observe the same event and construct entirely different interpretations of it, depending on the schemas active now of perception.

For NPC modelling, this framework has significant implications. A character that responds to player dialogue purely by retrieving pre-authored strings is not, in Neisser's terms, cognising. It is only reproducing. A credible cognitive model would require something structurally like a process in which the NPC generates an internal interpretation of what has been said, rather than simply matching input to output. The PNE's thought-pattern template system is designed around this intent. When a player delivers a dialogue choice, the NPC's cognitive layer does not retrieve a response; it generates a subjective interpretation, a belief, by selecting and instantiating from a library of thought patterns whose selection is dictated by the NPC's personality parameters. The result is that NPCs with different cognitive profiles generate different internal responses to identical player input, mirroring Neisser's central claim that interpretation is schema-driven rather than stimulus-determined.

However, the PNE's template matching is a deliberate simplification of Neisser's full model: it captures the essential output of constructive cognition; a generated interpretation rather than a retrieved response without attempting to replicate the underlying mechanism. The thought patterns produce plausible outputs, but the selection process is heuristic rather than formally cognitive.

3.2.2 Belief, Emotion, and the A-B-C Model: Ellis (1962)

Ellis (1962) introduced the foundational model of Rational Emotive Behaviour Therapy REBT): the A-B-C framework, in which an Activating event (A) does not directly produce an emotional Consequence (C), but does so only via the mediating Belief (B) that an individual holds about the event.

The same activating event produces entirely different emotional outcomes depending on this mediating belief. Ellis catalogued a range of irrational belief patterns. These patterns include hostile attribution (assuming negative intent on the part of others), catastrophising (treating manageable setbacks as existential threats), and projection (attributing one's own negative states to others).

The A-B-C model is a direct fit for the PNE's cognitive process. Player dialogue input is the activating event; the NPC's template matching thought is the belief for interpreting the event; the desire state is the emotional result. Ellis's identification of irrational types of belief patterns is also the inspirator of the types of cognitive bias categories implemented in the PNE's thought-pattern library. Hostile attribution bias which is the most common type of bias employed in the PNE's template set is Ellis's irrational pattern applied to interpersonal perception.

Ellis's model is clinical and prescriptive in its original framing, concerned with identifying and correcting maladaptive belief patterns in human patients. Its application to NPC design requires a reframing: where Ellis treats irrational beliefs as pathologies to be remedied, the PNE treats them as design parameters that define the character of an NPC's cognitive style. A character configured with high hostile attribution bias does not have a disorder; they have a personality. This distinction is important for the ethical considerations addressed in Chapter 2, where it is noted that NPC personality parameters are design tools, not clinical representations.

3.2.3 Attribution Theory: Heider (1958)

Attribution theory was founded by Heider (1958) when he observed that everyday social actors provide explanations for the behaviour of others by attributing mental states to them: beliefs, desires, intentions and sentiments. Those attributions are not complex explanations; they are largely inferences made when someone is seen performing an action. Heider observed the difference between internal attribution (attributing behaviour to the person's intention or disposition) and external attribution (attributing it to the situation), finding the systematic over-attribution of behaviour to personality and under-attribution to situation, which other researchers formalised as the fundamental attribution error.

Attribution theory is foundational to the PNE's design in two ways. First, it provides justification for modelling NPCs through explicit mental-state representations. Players naturally assume there are beliefs, desires, and intentions to characters they encounter (as Heider's framework predicts) then an NPC system that maintains explicit representations of those mental states is likely to produce behaviour that players interpret as psychologically coherent. What players think the character is "thinking" matches what the system is computing, instead of being something players just imagine on top of pre-written behaviour.

Second, Heider's internal/external attribution distinction maps directly onto the PNE's *"locus_of_control"* personality parameter. An NPC with a high internal locus attributes player behaviour to the player's character and intent; an NPC with a high external locus attributes the same behaviour to circumstances. This produces systematically different belief updates from identical player input, which in turn produces different desire formations and social intention selections downstream in the pipeline.

Heider also identified the concept of balance. This is the psychological discomfort that arises when an individual's beliefs about relationships are inconsistent, which maps onto the tension the desire layer must resolve when the player's relational standing with the NPC conflicts with the content of their dialogue choice.

Heider's model is descriptive and not prescriptive. The PNE models only one level: the player's intention to communicate, but not the player's beliefs about NPC beliefs. Multi-level mentalising can be architecturally realised but is not yet considered by the system.

3.3 Social Psychology and Interpersonal Dynamics

3.3.1 Influence and Compliance: Cialdini (1984)

Cialdini (1984) identified six principles of social influence: reciprocity, commitment and consistency, social proof, liking, authority, and scarcity. These principles are not cognitive but moreso social heuristics, patterns of influence that exploit the shortcuts human decision-making relies on when evaluating whether to comply with a request. Cialdini's core insight is that the same request can succeed or fail depending on its rhetorical framing: compliance is determined not only by what is asked but by how it is positioned relative to these principles.

The PNE provides four Language Arts of player speech - authority, diplomacy, empathy, and manipulation based on Cialdini's principles. Authority maps to his authority principle, diplomacy to liking and reciprocity, as empathy to liking, manipulation to commitment, social proof and scarcity. This relationship provides theoretical legitimacy for the Language Arts system: the four skill dimensions do not represent arbitrary gamified dialogue flavours but put into practice empirically validated social influence theory.

The NPC's personality parameters govern how strongly each Language Art is weighted in the skill-check calculation, reflecting personal differences in susceptibility to different influence strategies.

A character with high suggestibility to authority is computationally easier to influence through high-authority dialogue choices. A character with high reciprocity weighting responds more strongly to empathy-coded approaches.

Cialdini's framework is empirically grounded in compliance research conducted primarily in real-world commercial and interpersonal settings, and its application to fictional NPC interactions requires an implicit mapping assumption: that players interpreting NPC responses will do so through the same naive social cognition that operates in real-world interaction. This assumption is supported by the broader social psychology literature on social cognition in narrative contexts but is not independently validated within this project.

3.3.2 Social Comparison and Self-Esteem: Festinger (1954) via Guimond (2006)

Next, Guimond (2006) contributes to the PNE with social comparison theory which was originally proposed by Festinger (1954). It demonstrates that self-evaluation operates between groups as well as individuals, with in-group favouritism and out-group hostility emerging from comparative self-assessment. Critically, Guimond positions self-esteem as a relational construct shaped by perceived standing relative to others, not a fixed personality trait.

In the PNE, Guimond's work underpins the conceptualisation of the "self_esteem" and "faction" personality traits. Self-esteem among the NPCs is not a trait but rather is comparative: how an NPC evaluates their self-esteem relative to the player, their faction mates, and the enemy. This is the type of comparison Guimond describes.
NPC behaviour that's based on a character's faction, like an NPC being more hostile or wary if a player not from their in-group approaches, relies on these intergroup comparison processes as well.
The *"conf_indep"* (conformity to independence scalar) is also comparative but at the level of group membership. An NPC who tends to evaluate themself relative to more powerful or dominant others is more likely to be conformist, and less likely to be independent in making decisions. This then impacts on downstream intention choices. Guimond's research then provides an explanation of these parameters as explicit usage of comparative social evaluation theory.

3.3.3 Impression Management: Goffman (1959)

Goffman (1959) proposed a dramaturgical model of social interaction: people perform identities rather than simply express them. Every social actor maintains a front-stage presentation, the self-displayed in interaction and a backstage self. The private self is hidden from others.

Performance is governed not by authentic expression but by impression management: the strategic regulation of self-presentation to influence how others perceive and respond to one's behaviour. Key to the dramaturgical model is the observation that social interaction is a collaborative performance in which all parties both enact and are audience to one another.

This model of Goffman's corresponds to the structure of the PNE's NPC personality. The system has a cognitive layer where the NPC's beliefs and understanding are private. There also is a social layer, the behavioural intentions it makes public in social interaction. This is exactly Goffman's front and backstage, programmed. An NPC whose social intentions differ from its cognitive state is performing in Goffman's terms: it is presenting a managed self that does not reflect its "true" cognitive state. This potential dissonance between the private and the public makes NPC behaviour interpretable as strategic, rather than reactive, behaviour. The lack of correspondence between the NPCs behavioural intentions such as *“Assert Dominance”, “Deflect with Humour”, “Defend Cause Passionately”* and the actual state of the NPC is impression management in Goffman's sense: performative gestures chosen for their likely effect on the audience.

Goffman's model is sociological and not psychological and doesn't address how impression management decisions are computed. The PNE accounts for this through the intention registry with intentions that represent a finite set of actions from which the NPC selects in response to BDI state. Goffman's notion of frame breaks, where an extended performance breaks down, translates to the PNE's rising confrontation threshold, where the NPC's behaviour changes as psychological pressure reaches configured levels.

3.4 BDI Agent Architectures

3.4.1 Foundations: Bratman, Rao and Georgeff

Bratman's (1987) account of practical reasoning provides the philosophical foundation for the BDI model. He proposed that human deliberation operates through three core attitudes: beliefs, which represent the world as it currently is; desires, which represent preferred states; and intentions, which are commitments that lend stability to goal-directed behaviour over time. Crucially, Bratman argued that intentions are not simply a product of beliefs and desires combined — they function independently, constraining further deliberation so that agents are not perpetually re-evaluating every course of action they have already committed to.

Rao and Georgeff (1995) took this philosophical account and made it computable. Their formalisation defined BDI agents using possible worlds semantics and modal logic, treating belief, desire, and intention as modal operators over world states, with explicit satisfaction conditions governing how these attitudes must relate for an agent to count as rational.

It is worth noting that BDI is not one fixed architecture. Georgeff et al. (1999) traced how the framework evolved from Bratman's original philosophy into a family of computational variants, each emphasising different aspects, belief revision in some cases, commitment structures or inter-agent communication in others. It depends on what the application demands. Their retrospective is also significant because it directly validates extending BDI to domains, like NPC conversation, that earlier implementations never addressed.

BDI's appeal for NPC modelling is straightforward: it maps onto how people already think about other minds. Heider (1958) observed that ordinary social actors naturally explain each other's behaviour through beliefs, desires, and intentions. An NPC built on BDI representation inherits this quality. Players can look at what an NPC does and make sense of it using the same reasoning they apply to real people.

Scripted dialogue trees don't offer this. Their logic is hidden, and behaviour often feels arbitrary as a result. BDI makes the internal workings legible, not because the player sees the architecture, but because the behaviour it produces follows logic humans already understand intuitively.

3.4.2 Practical Reasoning: Wooldridge (2000)

Wooldridge (2000) identifies three ways a BDI agent can handle commitment. Blind commitment means pursuing an intention regardless of what changes in the world. Single-minded commitment means pursuing it until either the goal is reached, or it becomes clear the goal cannot be. Open-minded commitment is the most flexible — an intention is dropped as soon as it no longer fits what the agent wants.

The PNE uses a version of this third approach. After each player turn, the NPC recalculates its desire state from updated beliefs. If the new desire state doesn't support the prior intention, that intention is discarded rather than carried forward.

The practical effect is that NPCs don't get stuck. They respond to how the conversation is developing, rather than persisting in goals that have stopped making sense.

3.4.3 Practical Implementation: Bordini, Hübner and Wooldridge (2007)

Bordini, Hübner and Wooldridge (2007) produced what is probably the most practically influential translation of BDI theory into working code. Their AgentSpeak language defines agents through two core components: a belief base and a plan library. When a change in belief triggers a goal, the agent searches its available plans and selects one to execute. The Jason interpreter extends this foundation with goal failure handling, multi-agent communication, and various commitment strategies.

Jason works well in structured environments. Multi-agent simulation, robotic coordination, navigation and combat AI — these are domains where goal-directed behaviour operates within a manageable and largely predictable space. Conversation is different. The range of things a player might say cannot be enumerated in advance, and a pure AgentSpeak implementation would require a distinct plan for every possible input type. At any realistic conversational scope, the plan library becomes unworkable.

This is the central limitation the PNE is designed around. Rather than listing conversational plans, it uses template-matched thought patterns to handle the variability of player input at a higher level of abstraction. Each player turn produces a categorised internal response, encoding the NPC's interpretation, emotional register, and updated BDI state. Then the downstream pipeline can act on without needing a pre-authored plan for every piece of dialogue. Surface text generation is then handled by the LLM, working from the social intention that emerges from this process.

The result is a hybrid: BDI's reasoning structure is preserved, but the combinatorial authoring burden that Jason would impose is avoided (Bordini, Hübner and Wooldridge, 2007).

3.5 Interactive Narrative and NPC Dialogue Systems

3.5.1 The Problem of Agency: Murray (1997)

Murray (1997) identifies three properties that make digital environments function as narrative spaces: agency, immersion, and transformation. Agency is the player's capacity to take meaningful action and see it reflected in the story. Immersion is the sense of being inside the world. Transformation is the world changing in response to what the player does.

The tension she identifies is fundamental. Authored narrative requires authorial control. Player agency requires freedom which includes the freedom to do things the author didn't anticipate. These pull against each other. The tighter the author's grip on the story, the less real the player's choices become. Murray (1997) frames this as the central problem of interactive narrative: how to preserve craft without making agency an illusion.

Dialogue trees resolve this tension by giving up on agency entirely. The player picks from pre-written options; the NPC delivers a pre-written response. There is the appearance of choice, but nothing the player does can produce a reaction the author hasn't already scripted. This is apparent in criticisms of works such as The Walking Dead (Telltale Games, 2012) which the storyline has a multitude of options but little variation in the ending to each episode.

The PNE takes the opposite position. Players have genuine conversational freedom — they can say what they like — but authorial control is maintained at the level of structure rather than surface text. The author defines the NPC's personality, the scenario's shape, and the conditions that determine outcomes. The specific words spoken are generated, not written.

Murray's three properties map directly onto the architecture. Agency is delivered through the skill-check system. Immersion comes from LLM-generated dialogue grounded in the NPC's live

BDI state. Transformation is handled by the judgement-score-driven FSM, which routes the narrative based on accumulated conversational outcomes rather than predetermined triggers. The FSM is, in a real sense, Murray's (1997) practical answer to the authorship-agency problem — players shape outcomes, authors shape structure.

3.5.2 Emergent Narrative: Aylett (1999)

Aylett (1999) proposed emergent narrative as a way out of the conflict between scripted story and player freedom. Rather than pre-authoring narrative, the idea is to let it arise from the bottom up. From characters with their own internal states and goals, interacting with each other and with the player. She drew a distinction between cognitively determined behaviour, which is reasoned and goal-directed, and reactively determined behaviour, which is simply sensor-driven response. Believable interactive narrative, she argued, needs characters capable of the first.

She also introduced the concept of social presence — an extension of physical presence into social space, where characters communicate appropriate behaviour through social convention rather than explicit instruction. This matters because it offers a way to integrate player behaviour into emergent narrative without pre-scripting what the player is supposed to do.
The PNE's design rests on a similar premise. An NPC with a coherent internal model of beliefs, desires, and social intentions will produce behaviour that reads as socially present — not because its dialogue has been carefully authored, but because its responses follow a traceable internal logic. This is the system's answer to Murray's (1997) immersion property. Immersion here doesn't come from polished prose; it comes from consistency between what the NPC does and why.

Aylett (1999) is also honest about the risk. Emergent narrative might not emerge. The unpredictability that makes the approach interesting is the same quality that makes it fragile. There's no guarantee the conditions for narrative coherence will arise in any given interaction. The PNE's FSM addresses this directly. The judgement-score mechanism aggregates turn-by-turn conversational outcomes into a single scalar that drives narrative routing. Authors specify the conditions for state transitions, so meaningful story-level events occur even when individual exchanges are unpredictable. In Aylett's (1999) terms, the FSM is the shaping hand — it ensures that emergent interactions produce coherent outcomes without scripting what those interactions must be.

3.5.3 Drama Management: Mateas and Stern (2003)

Façade (Mateas and Stern, 2003) is the most technically acclaimed prior attempt to build BDI-grounded interactive drama. The system featured two NPC characters whose emotional and relational states were continuously updated by a drama manager responding to player input. Story coherence was maintained through a hierarchical task network, while the characters' immediate behaviour was driven by their internal emotional state. Player input was classified by a natural language understanding system into a vocabulary of dramatic acts.

It worked, and it showed the costs of working. Development took years. The authored behaviour library was enormous. And the system still broke down visibly when player input exceeded what the Natural Language Understanding could handle. When the drama manager's goals conflicted with the NPC state model, behaviour degraded. Mateas and Stern (2003) attributed this to the fundamental difficulty of reconciling bottom-up character agency with top-down narrative control. The PNE approaches this problem differently. It doesn't attempt natural language understanding at all. Players select from structured dialogue options that arrive pre-labelled with tone weights and skill requirements, so the system never has to parse free text. NPC responses emerge from the BDI pipeline without that classification step.

This is a narrower interaction model since players can't type whatever they like. But it's a more stable one. Narrative coherence isn't enforced by a drama manager overriding NPC state; it emerges from the FSM operating on accumulated judgement scores. The story stays on track because the aggregate of BDI outcomes naturally drives it forward.

3.6 Large Language Models in Game Contexts

3.6.1 Generative Agents: Park et al. (2023)

Park et al. (2023) populated a virtual town with twenty-five agents, each running on GPT-4 with a memory stream, reflection mechanism, and planning module. The agents formed relationships, attended events, and spread information. Behaviour that observers rated as humanlike and socially coherent. As a proof of concept, it was significant: LLM-driven social simulation at a scale and quality that hadn't been achieved before.

But the architecture rests on assumptions the PNE was built to avoid. In Park et al.'s (2023) system, the same model handles planning, reflection, and dialogue. Reasoning and generation are fully coupled. If the LLM produces an implausible plan, nothing catches it — there is no external mechanism to verify or correct the output. The system is also cloud-dependent, which makes it incompatible with offline consumer game deployment regardless of its other qualities.

The PNE separates these concerns. The BDI pipeline handles reasoning through deterministic template matching, with no LLM involvement at that stage. The LLM enters only at the point of surface text generation, working from the pipeline's output rather than producing its own plans. This means NPC behaviour stays coherent and author-controllable even when the LLM's output is imperfect. The NPC's internal state doesn't depend on the model getting things right because the model was never in charge of the state to begin with.

3.7 Synthesis: The Gap the PNE Fills

The literature reviewed in this chapter converges on a gap no prior system has filled in a form suited to commercial game development. It can be stated precisely: there is no existing middleware architecture that grounds NPC behaviour in principled cognitive and social psychology, maintains deterministic author-controllable narrative structure, uses locally deployable LLM inference for dialogue generation, and remains accessible to developers without the resources for large-scale dialogue authoring.

Each body of work gets part of the way there. Cognitive psychology — Neisser (1967), Ellis (1962), Heider (1958) — justifies mental-state-based NPC modelling but doesn't specify how to compute it. Social psychology — Cialdini (1984), Guimond (2006), Goffman (1959) — maps naturally onto NPC personality representation but has never been formalised in an interactive narrative context. BDI architectures, from Rao and Georgeff (1995) through Wooldridge (2000) to Bordini, Hübner and Wooldridge (2007), provide the computational substrate but have not been applied to conversational NPC systems with LLM generation. Narrative design research — Murray (1997), Aylett (1999), Mateas and Stern (2003) — defines the design problem clearly but produces either theoretical frameworks or research prototypes too costly and specialised for commercial use. Park et al. (2023) demonstrate LLM-driven social simulation at genuine scale, but in a form that requires cloud infrastructure and sacrifices the deterministic narrative control that game developers need.

The PNE sits at the intersection of these streams. Its BDI pipeline is grounded in the cognitive and social psychology literature. Its FSM provides the narrative scaffolding Aylett (1999) identified as necessary to stop emergent behaviour from collapsing into incoherence. Its LLM is constrained where Park et al.'s (2023) is not — author control is preserved, natural language generation is still possible. And its deployment model, local and engine-agnostic, targets the practical gap that all the above have left open.

Table 3.7 — Prior Systems Comparison (see Appendix I)

Property	**PNE (this work)**	**Façade (Mateas & Stern, 2003)**	**Generative Agents (Park et al., 2023)**	**Jason BDI (Bordini et al., 2007)**
Architecture	BDI pipeline + constrained LLM text generation	Drama-managed NLU + behaviour tree	LLM-driven memory, reflection, and planning	AgentSpeak interpreter; formal BDI goal stack
Personality model	6-dimensional continuous parameter space (cognitive + social) with wildcard	None — authored beat graph	Seeded natural-language persona description	Logical belief base; no continuous dimensions

Property	PNE (this work)	Façade (Mateas & Stern, 2003)	Generative Agents (Park et al., 2023)	Jason BDI (Bordini et al., 2007)
Reasoning layer	Deterministic (CognitiveThought Matcher); author-controllable	Deterministic drama manager	LLM performs reasoning and generation in one call	Deterministic AgentSpeak interpreter
Text generation	Local LLM (Ollama); constrained by closed intention registry + anchors	Rule-based NLG from authored templates	Cloud-hosted GPT-4; unconstrained	None — action-selection language only
Offline deployment	Yes	Yes	No — requires cloud API	Yes
Narrative control	FSM + judgement score; full author override; intention registry as design document	High — drama manager controls arc explicitly	Low — emergent from LLM; no guaranteed structure	High for task environments; not designed for open narrative
Developer access	JSON profiles + HTML creator; REST API; no pipeline code required	Very high authoring cost — years per experience	Requires cloud key + prompt engineering	Requires AgentSpeak and formal methods background
Emergent narrative	NPC side — personality parameters diverge outcomes across profiles	Drama-managed; emergence controlled by author	Agent side — social behaviour from LLM interaction	Not applicable — task-environment agents
Evaluation method	Structured log analysis + internal state inspection	User studies; qualitative player experience	Human evaluation of social plausibility	Formal verification; task-completion benchmarks

This is not to claim that the PNE is without prior art; it is to claim that the specific combination of properties it implements has not previously been assembled in a form suited to the target use case. The following chapters document the requirements analysis, design decisions, and implementation choices through which that combination was realised.

3.8 Summary

This chapter has drawn on four bodies of literature, each contributing something distinct to the PNE's design. Cognitive psychology from Neisser (1967), Ellis (1962), Heider (1958) grounds NPC behaviour in mental-state modelling and supplies the interpretive logic connecting player input to NPC response.

Social psychology from Cialdini (1984), Guimond (2006), Goffman (1959) provides the structural account of interpersonal dynamics that the personality model operationalises.

BDI agent architecture from Rao and Georgeff (1995), Wooldridge (2000), Bordini, Hübner and Wooldridge (2007) explores computational frameworks for deliberative NPC cognition.

Finally Interactive narrative research from Murray (1997), Aylett (1999), Mateas and Stern (2003) defines the core design tension: authorial control against player agency.

What none of these bodies of work provides, individually, is a path to practical implementation. The recent availability of locally deployable LLMs changes that. It is the condition that makes the architecture described in the chapters that follow buildable.

Chapter 4: Methodology

4.1 Data Collection

Data collection was qualitative and iterative. Rather than user studies or controlled experiments, the methodology treated the engine's own output logs as the primary source of evaluative evidence. Each test conversation was automatically serialised to a timestamped JSON file — for example, *amourie_oth0204.json*, a record of a four-turn exchanges between the player and the character *Amourie Othella* in the *door_guard_night* scenario. These logs captured every layer of the BDI pipeline's execution: the NPC's internal thought, subjective belief, emotional valence, desire state, behavioural intention, interaction outcome, and final attribute values, alongside the player's choice text, language art classification, and dice check results for each turn.

The purpose was diagnostic transparency rather than measurement. Reading each turn's internal pipeline state alongside the NPC's spoken response made dissonances directly legible. Cases where the generated dialogue was inconsistent with the pipeline's reasoning, or where NPC attributes drifted implausibly between turns.

The collection process was trial-and-error in character. A scenario was designed, the engine was run with a defined set of player choices, and the resulting JSON was examined — first by the eye, then with the assistance of an AI language model (Claude’s Opus Model, Anthropic) to identify narrative dissonance or technical faults.

Dissonance was defined qualitatively: any case where NPC response was incongruent with internal state, where an outcome label didn't match the response content, or where final attribute values seemed implausible given the choices made.

Technical faults included malformed JSON keys, missing terminal outcome fields, and desire states that matched no entry in the intention registry. The logs did not support formal statistical analysis; sample sizes were too small, conditions too variable across runs. Quantitative data was incidentally present in the form of numeric attribute values, and these were used informally to confirm that deltas were accumulating in the expected direction. No formal analysis of these values was conducted at this stage, that forms part of the Results and Discussion chapters.

One quantitative observation did directly drive a design revision. Early builds used a live LLM call to generate the NPC's internal thought and subjective belief on every turn. Logs from this phase recorded generation times of roughly 4–8 seconds per pipeline invocation, too lengthy for a real-time game context. This finding drove the decision, documented in the CHANGELOG entry for 9 March 2026, to replace the LLM-based cognitive layer with a deterministic template-matching system. Processing time dropped to under a second, with no reduction in the qualitative coherence of NPC thought outputs as assessed by log inspection.

4.2 System Design

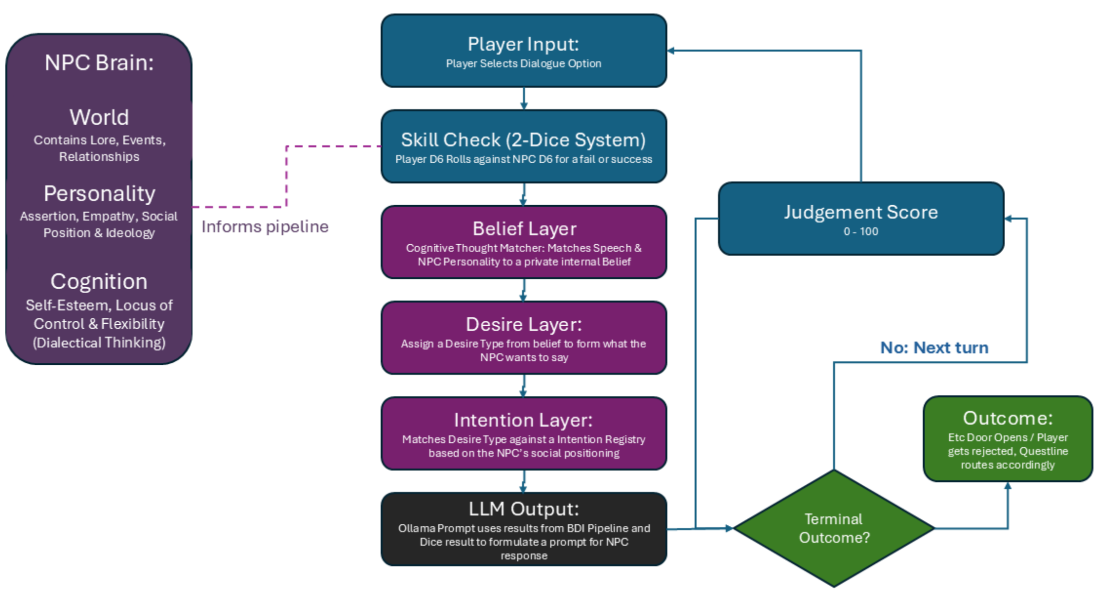


Figure 4.2 — PNE Pipeline Flowchart

The PNE implements a Belief-Desire-Intention (BDI) agent architecture (Bratman, 1987; Rao and Georgeff, 1995) for NPC behaviour in narrative game dialogue. Each player turn passes through five sequential stages — skill check, belief formation, desire formation, intention selection, and LLM output — with the NPC Brain informing the pipeline throughout. The Brain is a structure storing three components: World (lore, relationships, events), Personality (assertion, empathy, social position, ideology), and Cognition (self-esteem, locus of control, dialectical flexibility). These parameters remain stable across turns; they shape how each stage interprets player input rather than changing in response to it.

Execution begins when the player selects a dialogue option. A two-dice skill check resolves first, producing a success or failure that carries through the rest of the pipeline. The belief layer then matches the player's speech act against NPC personality to generate a private internal belief — what the NPC thinks. That belief drives desire formation, which assigns a desire type representing what the NPC wants to communicate. The intention layer maps that desire type against an intention registry filtered by the NPC's social positioning, selecting a concrete communicative intention. The LLM then receives the full pipeline output as a structured prompt and generates the NPC's spoken response via Ollama.

Each turn ends with a terminal outcome check. The judgement score — a running scalar between 0 and 100 — aggregates turn-by-turn results and determines whether the scenario resolves or the loop returns to player input.

4.2.1 Player Input

The point of entry into the BDI pipeline is the *PlayerDialogueInput* dataclass, defined in player_input.py. This structure encapsulates every attribute of the player's chosen dialogue option that subsequent pipeline stages require and is passed unchanged through all four layers so that each has access to an identical snapshot of the player's communicative act.

The most structurally significant attribute of a *PlayerDialogueInput* is its *language_art*, an enum drawn from the set *{CHALLENGE, DIPLOMATIC, EMPATHETIC, MANIPULATIVE, NEUTRAL}*. The language art classification determines which of the player's four skill dimensions is used for the dice check, and provides the primary categorical input to the cognitive thought-matching algorithm. Each scenario choice definition carries a pre-authored language art label; this is a design decision that keeps the categorisation deterministic and author-controlled rather than inferred from natural language.

The four language arts map directly to four player skill dimensions — authority, diplomacy, empathy, manipulation — encoded in the *PlayerSkillSet* dataclass. Each skill is an integer on a 0–10 scale. Skills are not general-purpose attributes of the player character; they govern the rhetorical register through which the player communicates. A player with high empathy skill is better at emotionally resonant dialogue choices; a player with high authority skill is better at commanding or challenging ones. This design is grounded in Cialdini's (1984) taxonomy of social influence strategies, as discussed in Chapter 3.

Each *PlayerDialogueInput* also carries four continuous tone signals — *authority_tone*, *diplomacy_tone*, *empathy_tone*, and *manipulation_tone*, each normalised to [0.0, 1.0] — encoded in the scenario choice definition. These are distinct from the language art classification: where the language art identifies which skill is checked, the tone signals convey the rhetorical intensity of each register within the choice. A choice classified as DIPLOMATIC may carry a moderate *authority_tone* if the diplomatic appeal is also assertive. Tone signals are consumed by the cognitive layer's emotional valence computation and by the cognitive thought template matching algorithm.

An optional *ideology_alignment* field identifies whether a choice appeals to a specific ideological position (e.g. "Communitarianism" or "Pragmatism"). This is matched against the NPC's *social.ideology* dict in the desire formation layer, enabling choices that invoke shared values to produce qualitatively different desire states from choices that do not, independently of the player's skill.

The skill check itself is implemented in the *SkillCheckSystem* class (skill_check.py). Two complementary mechanisms are present. The primary mechanism is a two-dice system: both player and NPC roll a single biased d6, and *player*_die >= *npc*_die constitutes a success. The player die is biased by normalising the relevant skill value to [0.0, 1.0] and using this as the exponential weight parameter in a weighted probability distribution over faces 1–6; the NPC die is biased by a derived resistance threshold calculated from the NPC's personality attributes. The resistance threshold for authority challenges, for example, is 0.3 + (social.*assertion* × 0.4): a highly assertive NPC resists commanding approaches more strongly. A secondary, legacy threshold-based mechanism is also retained, used internally to gate temporary NPC attribute modifiers — cases where a successful check temporarily shifts an NPC attribute value within the conversation.

This dual-mechanism design reflects the iterative development process. The threshold-based mechanism was the original implementation; the two-dice system was introduced later to give each outcome a degree of stochastic uncertainty that the player could observe and reason about, while still making probabilities analytically precomputable. The *success_probability*() method computes the pre-roll success percentage analytically — by summing the joint probabilities of all (*player_die*, *npc_die*) pairs where the player succeeds — so that the UI can display an estimated chance of success before the player commits to a choice, without requiring a simulation. A configurable difficulty modifier (SIMPLE: +0.15, STANDARD: 0.0, STRICT: -0.15) is applied as an additive bias adjustment on top of the player skill value, allowing the game to tune difficulty without modifying NPC profiles.

Table 4.2.1 — NPC Resistance Thresholds by Skill Dimension

Skill Dimension	NPC Attribute	Formula	Range (attr: 0 → 1)	Psychological Basis
Authority	social.assertion	0.3 + (assertion × 0.4)	0.30 → 0.70	Assertive NPCs resist being commanded; low-assertion NPCs defer to apparent authority
Manipulation	cognitive.self_esteem	0.2 + (self_esteem × 0.5)	0.20 → 0.70	High self-esteem NPCs are not susceptible to flattery or deception
Empathy	social.empathy	0.4 − (empathy × 0.2)	0.40 → 0.20	High-empathy NPCs are more receptive to emotional appeals — resistance decreases
Diplomacy	cognitive.cog_flexibility	0.3 − (cog_flexibility × 0.3)	0.30 → 0.00	Flexible thinkers are persuadable through reasoned argument — resistance approaches zero
(default)	—	0.5	0.50	Neutral fallback for unrecognised skill dimensions

4.2.2 Cognitive Layer

The cognitive layer is the first stage of the BDI pipeline. It receives the PlayerDialogueInput and the current NPC model, and produces a ThoughtReaction: a structured representation of the NPC's internal, unspoken response to what the player has said.

The *ThoughtReaction* consists of four components: *internal_thought* (a first-person reaction text), *subjective_belief* (the NPC's conscious interpretation of the player's communicative intent), *cognitive_state* (a snapshot of the NPC's three core cognitive attributes at the time of the turn), and *emotional_valence* (a scalar in [-1.0, 1.0] quantifying the NPC's affective response to the player's input).

The theoretical basis for this architecture is Ellis's (1962) A-B-C model of cognitive-emotional response, reviewed in Chapter 3. The player's choice constitutes the activating event (A); the *subjective_belief* output constitutes the belief (B) through which that event is interpreted; and the *emotional_valence* constitutes the emotional consequence (C). This explicit separation between stimulus and response, mediated by an interpretive belief state, is the design principle that distinguishes the PNE's NPC cognition from input-output scripting. Additionally, the cognitive bias categories implemented in the thought-template library draw on Beck's (1979) taxonomy of cognitive distortions, particularly his identification of automatic negative thoughts as systematic interpretive errors. Bias types such as hostile attribution, catastrophising, and black-and-white thinking are directly instantiated as template categories in the system.

The original implementation of the cognitive layer used a live LLM call — initially qwen2.5:3b, later phi3:mini — to generate the *internal_thought* and *subjective_belief* fields free-form, given a prompt containing the player's choice text and the NPC's personality parameters. Log inspection during the testing phase, described in Section 4.1, revealed two problems with this approach. First, response latency was prohibitive, typically 4–8 seconds per call on the available hardware, making real-time game integration impractical. Second, the free-form outputs were qualitatively inconsistent: the same player input and NPC configuration could produce structurally different thought formats across runs, and the subjective belief text was often too abstract to serve as reliable keyword input for the downstream desire formation layer. These issues were identified through systematic examination of output logs such as amourie_oth0204.json, where the NPC's belief-driven desire outputs showed poor correspondence between the stated belief and the desire category selected.

The replacement architecture, introduced on 9 March 2026, uses the *CognitiveThoughtMatcher* class (cognitive_thought_matcher.py) to perform deterministic template matching against a library of 810 pre-authored thought-pattern templates stored in *cognitive_thoughts.json*. Each template defines a *bias_type* — for example, *hostile_attribution*, *empathy_resonance*, or *cynical_realism* — alongside a set of *thought_variants* and *belief_variants*: text strings sampled randomly within the winning template to introduce surface variation without sacrificing determinism. A *match_weights* dictionary specifies the conditions under which each template should be selected.

The matching algorithm operates in three stages. First, every template is scored by evaluating its *match_weights* against the current *PlayerDialogueInput* and NPC state. Language art is treated as a discrete lookup: the player's current language art is located in the template's language art weight table, and the table's maximum value counts toward the total possible weight so that non-matching language arts are relatively penalised. Numeric parameters — including *npc_self_esteem*, *npc_locus_of_control*, *npc_cog_flexibility*, and the four tone signals — are evaluated as gate conditions: a parameter scores its defined weight if the extracted value falls within the template's specified range (using min, max, or both). The raw score is then normalised against the template's total possible weight to produce a fit score in [0.0, 1.0]. Second, the highest-scoring template above a threshold of 0.35 is selected as the winner. Third, if no template clears the threshold — indicating that the current input does not clearly match any template's profile — the system falls back to a *cynical_realism* default, which is designed to be contextually plausible across a wide range of inputs.

The *emotional_valence* computation is maintained as a separate rules-based calculation rather than a template output. This is because emotional valence is a continuous signal that needs to respond smoothly to continuous variation in tone scores, whereas the template matching produces a categorical output. The valence formula encodes four psychological relationships grounded in the literature: low self-esteem NPCs respond negatively to authority and manipulation (valence -= *authority_tone* × 0.3); external locus of control associates authority cues with perceived threat (valence -= *authority_tone* × 0.4); high cognitive flexibility responds positively to diplomacy and empathy; and rigid thinking resists persuasive tones. These relationships are direct implementations of the personality-behaviour mappings described in Ellis (1962), Heider (1958), and Guimond (2006), as reviewed in Chapter 3.

Table 4.2.2 — Emotional Valence Computation Rules (source: cognitive.py — _calculate_emotional_valence(); see Appendix J)

NPC Condition	Fires when…	Tone Signal	Effect	Formula	Psychological Basis
Low self-esteem	self_esteem < 0.4	Authority	Negative	− authority_tone × 0.3	Low self-esteem associates commanding tones with personal threat (Ellis, 1962)
Low self-esteem	self_esteem < 0.4	Manipulation	Negative	− manipulation_tone × 0.5	Susceptibility to manipulative framing; perceived exposure (Ellis, 1962)
External locus	locus_of_control < 0.5	Authority	Negative	− authority_tone × 0.4	External locus reads authority as

NPC Condition	Fires when…	Tone Signal	Effect	Formula	Psychological Basis
					situational coercion (Heider, 1958)
High flexibility	cog_flexibility > 0.6	Diplomacy	Positive	+ diplomacy_tone × 0.4	Flexible thinkers respond to reasoned, non-threatening framing (Guimond, 2006)
High flexibility	cog_flexibility > 0.6	Empathy	Positive	+ empathy_tone × 0.3	High flexibility correlates with empathic receptivity (Davis, 1983)
Rigid thinking	cog_flexibility < 0.4	Diplomacy	Negative	− diplomacy_tone × 0.2	Cognitively rigid NPCs resist persuasive framing (Ellis, 1962)

Rules are additive; multiple may fire on the same turn. Result is clamped to [−1.0, 1.0]. An NPC not meeting any condition receives a neutral valence of 0.0, driving desire formation to the long-term goal fallback.

4.2.3 Desire Layer

The desire layer is the second stage of the BDI pipeline, corresponding to the Desire component in the BDI architecture formalised by Rao and Georgeff (1995). It receives the ThoughtReaction from the cognitive layer, the PlayerDialogueInput, and the NPC model, and produces a DesireState: a structured representation of what the NPC wants in response to what it believes. The DesireState comprises three fields: immediate_desire (a natural-language description of the NPC's current goal), desire_type (a categorical motivational classification), and intensity (a scalar in [0.0, 1.0] indicating the strength of the desire).

The four desire types — information-seeking, affiliation, protection, and dominance — were derived from a review of the social psychology literature on interpersonal motivation and goal-directed behaviour. The information-seeking and affiliation categories correspond to Maslow's deficit needs in their interpersonal dimension and to Cialdini's (1984) framework of influence through liking and reciprocity; protection corresponds to threat-appraisal responses described in the social comparison literature (Guimond, 2006); dominance maps to Goffman's (1959) concept of face management under competitive social conditions. The deliberate restriction to four categories reflects a design trade-off: richer motivational taxonomies exist in the literature, but a larger category set reduces the reliability of downstream intention matching, since each desire type must be associated with a manageable set of intention templates in the intention registry.

The desire formation algorithm implements six belief-keyword patterns applied in priority order, followed by a bias modifier. The six patterns translate common features of the NPC's subjective_belief text into desire states:

1. **Uncertainty pattern** — keywords such as unclear, unsure, testing, and words in the belief text indicate that the NPC is not convinced by the player's communication, triggering information-seeking desire (or protection for low self-esteem NPCs).
2. **Sincerity pattern** — keywords such as *genuine*, *sincere*, and *authentic* indicate perceived authenticity, triggering affiliation (or guarded information-seeking for low-empathy NPCs).
3. **Threat pattern** — keywords such as *manipulative*, *threat*, and *deceive* indicate perceived hostile intent, triggering protection or dominance depending on the NPC's assertion level and wildcard configuration.
4. **Opportunism pattern** — keywords such as *opportunistic* and *exploit* indicate perceived self-interest, triggering information-seeking with elevated intensity.
5. **Ideology alignment pattern** — if the player's *ideology_alignment* tag matches a key in the NPC's *social.ideology* dict, the strength of that alignment determines whether affiliation or information-seeking is produced.
6. **Emotional valence fallback** — when no keyword pattern fires, negative valence (below -0.3) produces protection and positive valence (above 0.3) produces affiliation.

The pattern-matching mechanism for *subjective_belief* is the structural reason the cognitive layer must produce consistent, keyword-rich belief text rather than abstract prose. The original LLM-based cognitive layer failed this requirement because its outputs were too variable in phrasing, causing the desire formation step to fall through all six patterns to the valence fallback disproportionately often. This was directly observed in the output logs during the testing phase and is what prompted the switch to the template-based approach.

After the six patterns resolve a base desire state, a bias modifier is applied. The *BIAS_TO_DESIRE_MODIFIER* table maps each cognitive bias type (from the *CognitiveThoughtMatcher* output) to a potential override of *desire_type* and an additive boost to intensity. Hostile attribution bias, for example, always overrides desire type to protection and adds 0.2 intensity, this mechanism ensures that two NPCs with identical personality parameters, but different cognitive bias profiles will produce different desire states from the same player input, implementing the individual differences literature's account of how cognitive style mediates social perception.

Table 4.2.3 — BIAS_TO_DESIRE_MODIFIER

Bias Type	Desire Override	Intensity Boost	Effect
hostile_attribution	protection	+0.20	Ambiguous cues read as threatening regardless of surface content
optimism_bias	affiliation	+0.15	Sees opportunity in almost any input; gravitates toward connection
confirmation_bias	information-seeking	+0.10	Seeks validation of existing beliefs; mild boost to scrutiny
empathy_resonance	affiliation	+0.25	Unusually receptive to emotional signals; strong push toward connection (Davis, 1983)
cynical_realism	(no override)	+0.00	Accepts base desire without distortion; no bias amplification
ideological_filter	(no override)	+0.15	Frames input through ideology; intensifies the existing desire without redirecting it
self_referential	dominance	+0.10	Makes it personal; pushes toward self-assertion and face management
projection	protection	+0.10	Assumes the player wants what the NPC

Bias Type	Desire Override	Intensity Boost	Effect
			fears; pre-emptive defensiveness
in_group_bias	(no override)	+0.20	"Are they one of us?" — intensity boost without type change
black_white_thinking	dominance	+0.30	No middle ground; polarises toward aggressive self-assertion
scarcity_mindset	protection	+0.25	Fears loss above all else; strong protective push

4.2.4 Intention Layer

The intention layer is the third stage of the BDI pipeline, corresponding to the Intention component in Bratman's (1987) BDI formalism. It receives the DesireState from the desire layer and produces a *BehaviouralIntention*: a structured characterisation of how the NPC will behave in its spoken response. The *BehaviouralIntention* comprises an *intention_type* drawn from a closed canonical vocabulary, a *confrontation_level* in [0.0, 1.0], an *emotional_expression* acting direction, and a *boolean wildcard_triggered* flag.

The layer is implemented by the *SocialisationFilter* class (social.py), which selects the best-matching template from the INTENTION_REGISTRY — a statically defined list of *IntentionTemplate* instances (intention_registry.py). Every template specifies a *desire_type* it serves, a list of *desire_keywords* that activate it, a valid confrontation range, an emotional expression label, and optional hard gates: *wildcard_required* restricts the template to NPCs with a specific wildcard trait, and *npc_conditions* specifies attribute thresholds that must be met. This makes the full set of NPC behavioural possibilities author-enumerable — the intention registry is a design document as much as a code artefact.

Selection mirrors the thought matcher's structure but operates categorically. The registry is first filtered to templates whose *desire_type* matches the current desire state. Each surviving candidate is then scored on three criteria: keyword overlap with the desire text (up to 0.5), confrontation band fit with partial credit for near misses (up to 0.4), and an intensity bonus of 0.1 for high-intensity desires matched to high-confrontation templates. Hard gates are evaluated before scoring — templates requiring a wildcard the NPC lacks, or specifying attribute conditions the NPC doesn't meet, are excluded with a score of -1.0. The highest-scoring template is selected.

The registry was a significant architectural revision (CHANGELOG, 2 March 2026). The original design allowed the socialisation layer to return free-form intention strings, passed directly to the LLM prompt and to the scenario FSM as transition conditions. This created a reliability failure: FSM transitions could not dependably match LLM-generated intention text because phrasing varied across runs. Replacing free-form generation with a closed canonical vocabulary made routing deterministic — every intention type reaching the scenario layer is guaranteed to be a registered name, and scenario authors can write transitions against those names with confidence. A wildcard override mechanism bypasses the normal desire-to-intention flow for extreme personality configurations. The Inferiority wildcard hard-triggers the Submit intention whenever the player's authority tone exceeds 0.5, regardless of current desire state. The Napoleon wildcard enables Assert Dominance Aggressively — otherwise unreachable because its confrontation range of 0.8–1.0 exceeds what the normal scoring algorithm produces for typical NPC profiles. This ensures the full range of personality archetypes remains expressible at runtime.

The selected intention's confrontation level is not simply inherited from its template range. It is computed by clamping the NPC's natural confrontation tendency — derived as assertion × 0.7 + *conf_independence* × 0.3, where *conf_independence* represents the NPC's position on a conformity-to-independence spectrum, with higher values producing greater confrontational tendency — to the template's valid range, then nudging upward by a proportion of desire intensity. This produces a continuous confrontation value that varies within the template's range based on both personality and situational pressure, giving NPCs with the same intention type qualitatively different deliveries.

4.2.5 Outcome Layer

The outcome layer is the final stage of the BDI pipeline. It receives the *BehaviouralIntention* from the socialisation layer and maps it to two distinct output types: an *InteractionOutcome* that modifies NPC state and advances the conversation, and an optional *TerminalOutcome* that ends the conversation with a defined narrative consequence.

This two-tier structure was not present in the original design. The initial implementation (CHANGELOG, 22 November 2025) defined a single outcome type checked at every turn. Log inspection during testing revealed the problem: conversations ended abruptly, the relationship between accumulated player choices and the outcome was unclear, and the narrative unbounded. The two-tier system was the response.

An *InteractionOutcome* represents the micro-consequence of a single turn. It carries four components: *stance_delta*, a dictionary mapping NPC attribute paths to additive changes applied immediately to the NPC model; *relation_delta*, a float applied to *world.player_relation*; *intention_shift*, an optional string that replaces the NPC's current long-term intention; and two scripted response variants, *min_response* and *max_response*, that anchor LLM output at the extremes of the emotional valence scale. These variants act as scene direction rather than replacements. The LLM generates a response coherent with the current valence level, constrained by both anchors. This addressed the free-form quality degradation observed when no constraints were provided.

A *TerminalOutcome* represents the consequence of the full conversation. It carries a *terminal_id* enum value, a callable condition function evaluated against the current NPC and conversation state, a result string describing the narrative consequence, and a *final_dialogue* line. Terminal conditions are evaluated at the end of every turn by *OutcomeIndex.check_terminal_outcomes*(); the first condition that evaluates to true ends the conversation. Multiple terminal outcomes per scenario node are supported, each with its own condition function, enabling branching endpoints that respond to accumulated conversational state rather than just the most recent turn.

The judgement tracking system, introduced on 3 March 2026, was the direct precursor to this architecture. Terminal routing had previously been driven by a dysfunctional system. The judgement score replaces this with an integer initialised at 50 and shifted by each dice outcome, scaled by a risk multiplier when the pre-roll odds were against the player. Integrating evidence across turns this way creates a narrative arc that rewards sustained successful interaction and allows recovery from early failures, reflecting Murray's (1997) principle that meaningful agency requires choices to carry cumulative weight.

4.3 Approach Ethics

4.3.1 Ethical Approach

The primary ethical consideration in this project is the responsible use of psychological theory as a design resource. The NPC personality parameters such as cognitive flexibility, self-esteem, locus of control, assertion and empathy are drawn from established clinical and social psychology literature. Applying clinical constructs in a game design context raises two concerns: trivialising conditions that carry real-world significance and producing NPC behaviours that model or reinforce harmful social dynamics.

Both were addressed through deliberate constraint. Personality parameters are treated as design variables, not diagnostic categories. Clinical terminology does not appear in any player-facing output, the labels exist only in the developer-facing NPC profile JSON. A character with low self-esteem and high hostile attribution bias is communicated to the player entirely through behaviour, not framing.

No NPC in the test corpus was designed to model a real individual or clinical population. Characters are fictional inhabitants of a fictional post-collapse city, and their psychological profiles were constructed for narrative plausibility rather than clinical fidelity.

The use of Claude (Anthropic) during data collection — specifically to review output logs for narrative dissonance — is a methodological transparency concern worth stating plainly. The AI was used as a pattern-recognition aid to flag candidate issues in long JSON files. All observations were reviewed and critically evaluated personally before acting on them. This is AI-assisted tooling, not independent analysis, and falls within acceptable use guidelines.

All NPC characters and narrative content are original creations. No copyrighted fictional material was used in the construction of NPC profiles or scenario dialogue. Academic sources are cited in accordance with the Harvard referencing convention; no source material was reproduced without attribution.

Chapter 5: Design Analysis & Optimisation

5.1 Model Design

5.1.1 Overview of the Final Pipeline

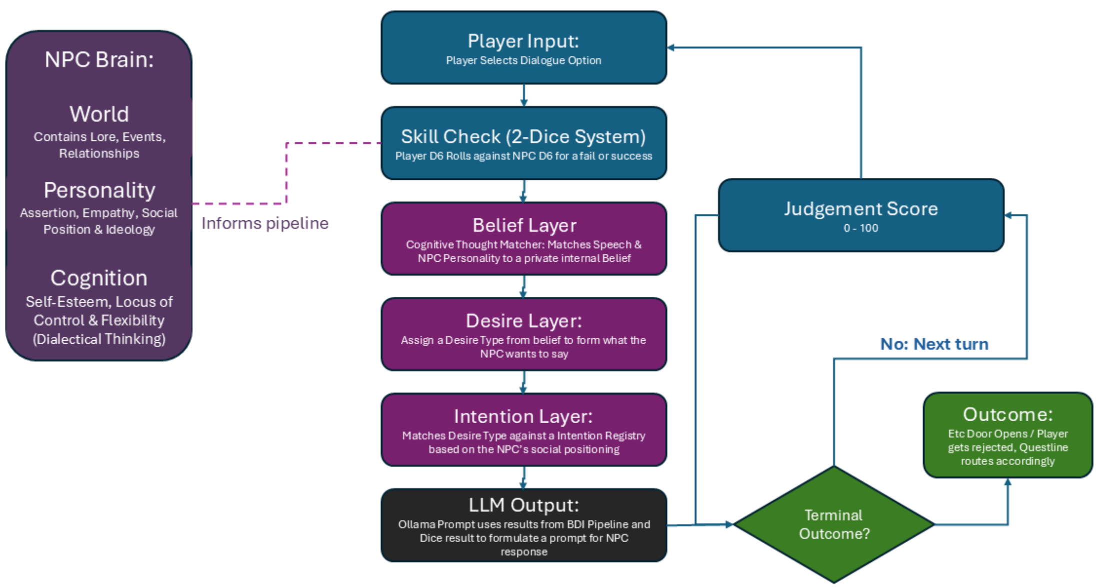


Figure 4.2 — PNE Pipeline Flowchart (stage-by-stage summary: see Appendix A, Table A.1).

5.1.2 Core vs. Supporting Components

The core components are the Player Input and Skill Check, the Cognitive layer (via *CognitiveThoughtMatcher*), Desire Layer, Socialisation/Intention Layer, and Outcome Layer.

These are what the PNE's claim to psychologically grounded NPC behaviour cannot be sustained without.

Supporting components (the FastAPI REST/WebSocket API layer, HTML character creator & world creator, and interactive dashboard) enhance usability but are not central to the theoretical argument.

5.1.3 Key Design Decisions

Language: Python and JSON

Python was selected due to the language having faster iteration and native JSON support. The use of JSON was due to it being language agnostic. This was a deliberate decision to make the PNE integrate with Unity (C#), Godot (GDScript), and Unreal Engine (C++). All of which provide native JSON parsing, meaning any game engine can consume NPC profile data and session logs without format translation. This portability was a primary motivation for structuring all persistent data as JSON rather than a database schema.

Local LLM via Ollama: *qwen2.5:3b*

Language generation was performed through a locally hosted model to avoid the issue of cloud models increasing latency, as gameplay requires responsiveness fundamentally. It also removes the requirement for an active internet connection for games that might be offline compatible.

Ollama runs the LLM as a local process with no external traffic, preserving player privacy and enabling offline operation.

qwen2.5:3b was selected to fit within 4 GB VRAM. A somewhat realistic minimum for a budget consumer PC, while producing coherent language generation. This is a key trade-off, prioritising accessibility over quality. The architecture is designed to improve transparently when a more capable model is substituted.

BDI Architecture

BDI was utilised because the cross-domain compatibility with psychology. Ellis's (1962) A-B-C model maps onto belief updating, Cialdini (1984) onto desire formation, and Goffman's (1959) impression management onto intention selection.

5.1.4 NPC Data Structure

Three sections define each NPC as a JSON object.

Cognitive; (*self_esteem*, *locus_of_control*, *cog_flexibility)*.

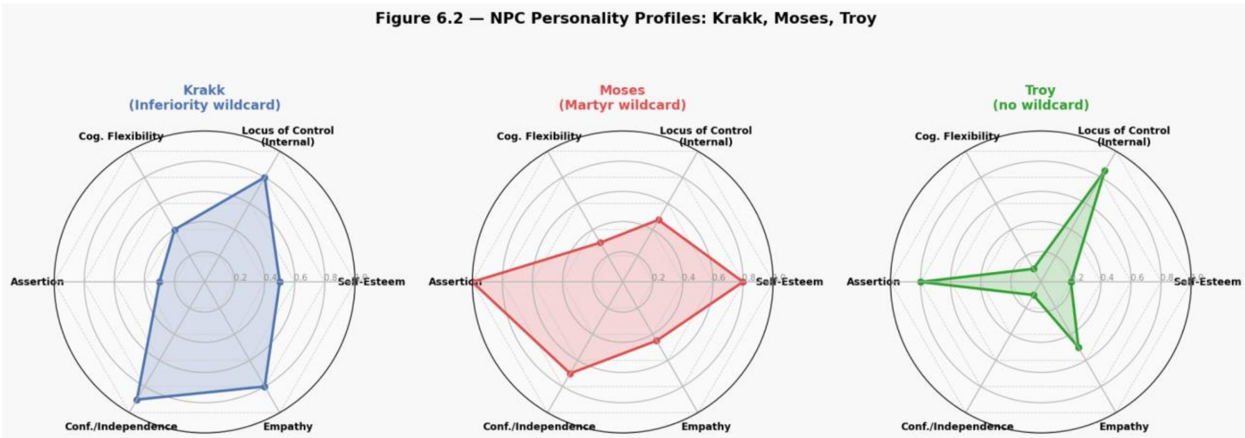
Social; (*assertion*, *conf_indep*, *empathy*, *ideology*, *wildcard*.)

World; (*personal_history*, *player_relation*, *faction*, *known_events*.) Everything normalised to [0.0, 1.0].

The benefit to this model is that a narrative designer doesn't need to touch the pipeline to author a character. NPCs with Higher assertion means more confrontational communication strategies. Higher empathy means the NPC responds better to sincerity. The wildcard field is the exception; it unlocks behavioural archetypes that the standard parameter space simply can't reach.

Figure 5.1.4 — NPC data structure. The three-block JSON schema (cognitive, social, world) is described in full in Section 5.1.4. Personality parameter profiles for Krakk, Moses, and Troy are visualised in Figure 6.2 (Appendix G).

The full character creator tool (*character_creator.html*) provides a form-based interface for authoring profiles without editing JSON directly.



Each chart plots the six core personality dimensions (Self-Esteem, Locus of Control, Cognitive Flexibility, Assertion, Conf./Independence, Empathy) on a normalised [0, 1] scale. The contrast between profiles illustrates how different parameter configurations encode structurally distinct psychological archetypes.

5.2 Interpretability

5.2.1 How Psychological State Influences LLM Output

The PNE does not allow the LLM to operate as a free-form text generator. The output of every prior pipeline stage is injected into the LLM prompt as structured scene direction, constraining what the NPC says without dictating exact text.

The pipeline's output to Ollama combines to provide a psychological context. From the cognitive layer: *internal_thought* and *subjective_belief* - the NPC's own thoughts and what it thinks of the player's motives. From the desire layer: *immediate_desire* and *desire_type* - its desires. From the intention layer: *intention_type* and *emotional_expression* - how it wants to express itself. From the outcome layer: *min_response* and *max_response* - the stylistic restrictions on how it should respond. And *emotional_valence* - the mood of the NPC, positive or negative.

The LLM is not to respond in any way to the player's input. It is to speak on behalf of an NPC with a given *emotional_valence*. The LLM is for phrasing, rhythm, register. The psychology is not. This is how the system solves the problem of unreliable open-ended dialogue with an LLM. We still get hallucination and drift - but in a limited context. And when the model follows the prompt, it generates a response that sounds like it's from the NFC's psychology - but it doesn't repeat itself.

5.2.2 Amourie Othella & The Door Guard Scenario

The following section traces a single four-turn conversation between the player and Amourie Othella (*door_guard_scenario*, log: *new_amourie_door.json*). Amourie's profile encodes high self-esteem (0.9), strongly internal locus of control (0.2), moderate cognitive flexibility (0.8), moderate assertion and empathy, a Communitarian ideology, and the Napoleon wildcard. Her starting *player_relation* is 0.5.

Turn 1: Authority Challenge

The player selects: *"I don't need to prove anything to you. I've fought for survival just like you have. Open the door."*

This choice carries the *CHALLENGE* language art, triggering the authority skill dimension in the dice check. The skill check resolves against Amourie's resistance threshold, which is elevated by her high assertion ($0.3 + 0.8 \times 0.4 = 0.62$). The check fails; the authority tone is high but so is her resistance.

The cognitive layer matches this input against the thought-template library. The high authority tone and the NPC's moderate cognitive flexibility score a template in the *cynical_realism* bias category. The *internal_thought* output is: *"Maybe their words are genuine, but I can't fully trust them."* The *subjective_belief* is: *"They claim empathy but might be using me."* The *emotional_valence* computation reflects the authority tone applied to a high-self-esteem, internal-locus NPC: rather than suppression, it produces moderate negative affect (-0.26), signalling perceived challenge rather than threat.

The desire layer evaluates the subjective belief for keyword patterns. The belief text contains the keyword using (proximate to the manipulative pattern), triggering a protection desire with elevated intensity (0.8). However, Amourie's Napoleon wildcard is checked: the wildcard does not fire here because the player's authority tone, while high, does not yet meet the wildcard override threshold. The desire resolves as information-seeking at intensity 0.8.

The intention layer selects "*Carefully Question Motives.*" A medium-confrontation template (*confrontation_level*: 0.46) consistent with the information-seeking desire and Amourie's personality profile. The confrontation value is computed by clamping Amourie's natural confrontation tendency to the template's valid range and pushing it upward proportionate to the desire intensity.

The outcome layer applies a *relation_delta* of -0.2, dropping the relation from 0.5 to 0.3. The LLM is given the state summary including *min_response*: *"You think you can intimidate ME? Wrong move."* and *max_response*: *"Bold. I respect that. But respect isn't enough."* With negative emotional valence, the output tends toward the min anchor. The generated response is: *"Bold. I respect that. But respect isn't enough."* It is acknowledged here that in this turn, the LLM produces the *max_response* rather than generating original text.

Turn 2 — Personal Sacrifice

The player switches register: *"I've already lost everything. My family, my home. I have nothing left but the will to fight."*

This choice carries the *EMPATHETIC* language art and high *empathy_tone*. The cognitive layer matches a template closer to *empathy_resonance*. The *internal_thought* is: *"These words cut deep, real or not?"*; the *subjective_belief* is: *"Player seems genuinely broken and seeking support."* The valence shifts to +0.27.

The desire layer finds the keyword genuine in the belief text (sincerity pattern), triggering an affiliation desire. Amourie's empathy score of 0.75 does not suppress this pattern. The desire resolves as information-seeking at moderate intensity (0.4), reflecting the guarded quality of a character who is moved but not yet convinced.

The intention layer selects Neutral Response with confrontation at 0.6, higher than turn 1's value despite the more empathetic player choice, because the desire intensity is lower and Amourie's base confrontation tendency (derived from her assertion and *conf_indep*) clamps the value upward. The outcome applies *relation_delta*: +0.25, recovering the relation from 0.3 to 0.55. The LLM generates: *"Then you understand. That loss... that rage... use it well."*

Turns 3–4 and Terminal Outcome

Subsequent turns accumulate positive relation deltas as the player reaffirms commitment and accepts Amourie's terms. By turn 4, *player_relation* has reached 0.95 and the judgement score has accumulated sufficient positive weight. The terminal outcome check evaluates the condition *player_relation* >= 0.9 and *judgement_score* >= 65, which passes. The conversation ends with the terminal outcome SUCCEED, and the scenario FSM routes to the succeed node.

Table 5.2.2 — Amourie Othella: Relation Trajectory Across Turns *(source: new_amourie_door.json)*

Turn	Player Choice	Outcome Intention	Δ Relation	Relation
Start				0.50
1	authority_challenge	Resist Player's Authority	**−0.20**	0.30
2	personal_sacrifice	Accept Player as Ally	**+0.25**	0.55

Turn	Player Choice	Outcome Intention	Δ Relation	Relation
3	reaffirm_commitment	Test Under Pressure	**+0.20**	0.75
4	accept_terms	Accept Player for Trial	**+0.20**	0.95
Terminal		**SUCCEED**		**0.95**

5.2.3 Comparative Output Analysis

Full BDI conversation breakdowns for Krakk Klikowicz, Morisson Moses, and Troy are provided in Appendices B–D. Comparative run analysis is presented in Section 6.5.

The qualitative prediction is clear from the architecture: a player with high *authority* skill and low empathy skill will produce systematically different pipeline states. It reliably produces different thought templates, different desire types, different intention selections, different relation deltas from a player whose stats invert those values, even if both players select identical choice text. The main engine fault is that Ollama may sometimes misfire and rely on min or max responses instead of generating original text.

The mechanism for this difference is the *PlayerSkillSet*, which biases the player's dice roll, and the *language_art* classification, which determines which NPC resistance formula is applied and which tone signals populate the cognitive layer inputs.

5.3 Optimisation

5.3.1 Architectural Evolution: From Prototype to Current System

The most significant architectural change between the initial prototype and the current system was the removal of the initial Ollama call in the cognitive layer and its replacement with the deterministic *CognitiveThoughtMatcher*.

Earlier in the prototype, every pipeline invocation made two LLM calls: one to generate the NPC's *internal_thought* and *subjective_belief*, and one to generate the final spoken response.

Generation times were approximately 4 to 8 seconds per PNE invocation. In a game context where player-facing response time directly determines the quality of the experience, this latency was a massive issue.

The *CognitiveThoughtMatcher*, introduced on 9 March 2026, replaced the first LLM call entirely. The matcher operates against a library of 810 pre-authored thought-pattern templates (*cognitive_thoughts.json*), each carrying a *bias_type,* variant text strings, and a *match_weights* dictionary. Template selection is a weighted scoring algorithm that executes in sub-millisecond time. The practical result was a reduction in cognitive processing from 4–8 seconds to 3 seconds at worst. with no material reduction in the qualitative coherence of thought outputs as assessed by log inspection.

The second major change was to replace the generation of unstructured intention strings with the *INTENTION_REGISTRY*. The prototype socialisation layer produced uncontrolled intention strings that were used directly as conditions to the scenario routing mechanism.

The non-deterministic nature of the LLM used to generate these strings meant that the PNE struggled to have a clear way of determining conversational direction resulting in routing problems that could be observed in the logs as unexpected ends of conversations or repeated nodes. Switching to a finite set of intention types instead made a deterministic FSM system possible. The scenario FSM always receives one of a familiar set of strings, and its transitions will always match.

The terminal outcome architecture was a third structural revision, introduced in response to narrative incoherence observed in early test logs. The prototype's single-outcome system allowed conversations to end at any turn, producing abrupt terminations with no narrative arc. The two-tier system (interaction outcomes + terminal outcomes) separates per-turn consequences from full-conversation consequences, and the judgement score, a 0–100 integer that aggregates dice outcomes across turns provides a more robust routing signal than the raw *player_relation* float, which proved too volatile under small per-turn deltas.

Table 5.3.1b — Pipeline Architecture: Prototype vs Current System (see Appendix H)

Stage	**Prototype**	**Current System**	**Change**
1. NPC Intent	No world context; personal history only	Full cognitive/social/world profile with known events, figures, faction.	World context added 2 Mar 2026 — NPCs respond to sociopolitical environment
2. Choice Selection	Free-text input, no tone decomposition	Structured input: language art + four tone scores + ideology alignment tag	Enables deterministic scoring downstream

Stage	Prototype	Current System	Change
2a. Skill Check	Threshold-only (player skill vs fixed value)	Two-dice system (biased d6 × d6) + analytical success_probability()	Stochastic uncertainty with pre-roll probability display for the UI
3. Cognitive Layer	LLM call #1 (~4–8 s) uncontrolled thought generation	**C**ognitiveThoughtMatcher + 810 templates, deterministic	Eliminated latency; produced keyword-rich belief text for desire layer
4. Desire Layer	2-type valence lookup	6 keyword patterns + BIAS_TO_DESIRE_MODIFIER (11 bias types, 4 desire types)	Richer desire differentiation: cognitive bias now shapes desire independently of valence
5. Socialisation	Free-form intention string & experimental FSM routing (unreliable matching)	INTENTION_REGISTRY. closed 19-type vocabulary + FSM (deterministic)	Fixed routing failures caused by LLM phrasing variation
6 Output	LLM call #2, unconstrained; single outcome tier	Only LLM call with min/max anchors; two-tier outcomes	Constrained generation; separated per-turn from scene-level consequences
7. Terminal	Raw *player_relation* float threshold	*judgement_score* (0–100 integer) + *player_relation* combined	Float too volatile; integer score gives stable arc-level routing signal

Metric	Prototype	Current
LLM calls per turn	**2**	**1**
Cognitive processing	4–8 s	0 ms
Intention vocabulary	Open / uncontrolled	Closed / 19 types
FSM routing	Non-deterministic	Deterministic

Table E.1 — Development Timeline: Major System Milestones (see Appendix E)

5.3.2 Parameter Tuning

The parameters controlling the pipeline were difficulty modifiers, skill check thresholds, resistance formula weights, bias intensity boosts and confrontation clamping values. These were tuned iteratively, sometimes with the help of AI (Claude Opus Model, Anthropic) after feeding it multiple output logs. The cycle was simple: test run, log inspection, value adjustment. The evaluation criterion throughout was qualitative: did the output reflect plausible NPC behaviour for the given personality profile and player input?

Difficulty modifier values *(SIMPLE: +0.15, STANDARD: 0.0, STRICT: -0.15)* were established by running each tier against multiple NPC profiles, calibrating until success rates felt appropriately challenging. Not trivial at simple, not impossible at strict. The resistance formula constants followed the same logic. The authority challenge formula *(0.3 + assertion × 0.4)* was tuned so that a maximally assertive NPC produces a resistance of 0.7, difficult but beatable, while a minimally assertive NPC produces 0.3, still non-trivial.

The *BIAS_TO_DESIRE_MODIFIER* intensity boosts required more careful calibration. Too large a boost effectively bypasses the keyword-pattern logic, too small renders the modifier invisible in the output. The target was a value that reliably shifts downstream desire consequences without overwhelming the pattern matching that precedes it. Claude was used as a diagnostic aid throughout. Output logs were provided alongside a description of expected behaviour, and discrepancies were used to isolate which parameter was out of range. Any case where that diagnosis was inconclusive was followed up with manual investigation.

5.3.3 Limitations

The Ollama Bottleneck

The LLM integration is the system's primary limitation and its most distinctive feature. It is what gives NPCs their linguistic expressiveness, and it is where the experimental nature of the engine is most apparent.

The qwen2.5:3b model, operating within a 4 GB VRAM budget (1650 Super), produces acceptable generation times at low concurrency. In a live game scenario with multiple simultaneous NPC conversations, that would degrade. More substantially, the model does not reliably adhere to prompt constraints across all inputs. Where the emotional context is complex or the *min_response*/*max_response* anchors sit far from the model's training distribution, responses can drift.

The model may produce output that is grammatically coherent but tonally inconsistent with the specified behavioural intention, or append meta-commentary on its own response, as observed in several turns of *new_amourie_door.json*. Additionally, if the Ollama model is to fail, it displays the min / max content rather than generated text. This is only a visible issue developer side, rather than something an end-user may notice, however.

This is a known property of small-scale quantised models. Instruction-following in qwen2.5:3b is materially weaker than in qwen2.5:7b or qwen2.5:14b, which would require 8 GB and 16 GB VRAM respectively. The system anticipates this. Because the pipeline produces a complete, structured psychological context before the LLM is ever called, a player on more capable hardware who substitutes a larger model receives the same deterministic pipeline behaviour with qualitatively better surface text. No modification to the engine required. That scalability is a design property, not an accident.

Audience Constraint

In its current form, the system is best suited to the mid-to-high end of the consumer PC market. Players on laptops with integrated graphics, or older discrete GPUs below the 4 GB VRAM threshold, cannot run a capable model in a playable configuration. This is a real constraint on potential audience, and it distinguishes the PNE from cloud-based dialogue systems, which impose no local hardware requirement at all. The trade-off is intentional; player privacy and offline operation are preserved. But it is still a restriction that must be acknowledged.

Prompt Adherence and Hallucination

The qwen2.5:3b model cannot generate quality dialogue text at the level of current state-of-the-art models. It sometimes hallucinates information not in the prompt, speaks about its own role, or is tonally off. These are evident in the test logs and differ in kind from the deterministic stages of the pipeline. More importantly, they're disentangled. Being independent of the LLM output, the NPC's internal state is always valid, even if the dialogue is poor. Upgrading the model improves the dialogue. Nothing else changes.

Scenario Scope

The test dataset is small; a few scenarios built around a door being guarded. The system has not been evaluated against conversations longer than four turns per node. This presents gaps in validation coverage but not design failures. It does however limit how confidently the system's behaviour can be generalised beyond the tested scenarios.

Chapter 6: Results

6.1 Overview

This chapter showcases results from test conversations with three NPCs in the *door_guard_night* scenario: Krystian 'Krakk' Klikowicz, Morisson Moses and Troy. Each has a different psychological profile, different cognitive parameters, social parameters, ideological parameters and (in two cases) wildcard factors.

Testing three profiles will determine whether the pipeline generates distinct internal states for the same player prompts, and that these differences are attributable to the NPC personality parameters, not LLM variability. For each NPC, we first show the entire JSON profile, then a turn-by-turn breakdown of the BDI profile stored in the conversation logs. A fourth set, two comparative tests against Troy with different player skill setups, are presented in Section 6.4. It has a specific purpose: does the distribution of player stats play out differently against the same NPC in the same scenario?

One important note is all the results in this chapter are based on system conversation logs. There was no user study, human evaluation or third-party review. This was by design; the engine's built-in output in structured JSON format makes it possible to inspect the output of every layer of the pipeline on every turn, which is diagnostic information that would be unavailable from a user study. This comes at the expense of being able to make claims about perceived believability of the NPC, naturalness of the conversation, or player experience. These are addressed in Chapter 7.

Full turn-by-turn BDI breakdowns for all three NPCs are recorded in Tables B.1 (Krakk), C.1 (Morisson Moses), and D.1 (Troy).

6.2 NPC Profiles

6.2.1 Krystian 'Krakk' Klikowicz

Table 6.2.1 — NPC JSON Profile: Krystian 'Krakk' Klikowicz

Block	**Field**	**Value**
Metadata	Name	Krystian 'Krakk' Klikowicz
	Age	28
	Faction	Runners
	Social Position	Boss

Block	Field	Value
Cognitive	self_esteem	0.5
	locus_of_control	0.8
	cog_flexibility	0.4
Social	assertion	0.3
	conf_indep	0.9
	empathy	0.8
	ideology	Libertarianism 0.9, Individualism 0.1
	wildcard	Inferiority
World	player_relation (start)	0.5
	known_events	hop_removal, dysphoria_expansion, runner_coordination
	known_figures	Amourie Othella, Morisson Moses, Jean Pope Chautlier

Krakk's profile is that of someone questioning yet non-confrontational. His low assertion (0.3) and high empathy (0.8) mean he is responsive to genuine, properly framed appeals; his very high conf_indep (0.9) means he is almost completely reliant on his personal judgement and is unlikely to be swayed by appeals from outside his own worldview. His high Libertarian ideology (0.9) makes him particularly responsive to messages that emphasise the freedom of movement, utility and independence of the player's offer. The Inferiority wildcard introduces a hard override under pressure - if the player's tone of authority exceeds a certain threshold, Krakk's desire to prove his superiority can overwhelm the normal intention flow, although this was not triggered in the test run here.

6.2.2 Morisson Moses

Table 6.2.2 — NPC JSON Profile: Morisson Moses

Block	Field	Value
Metadata	Name	Morisson Moses

Block	Field	Value
	Age	35
	Faction	Insurgency
	Social Position	Boss
Cognitive	self_esteem	0.8
	locus_of_control	0.475
	cog_flexibility	0.3
Social	assertion	1.0
	conf_indep	0.7
	empathy	0.45
	ideology	Utilitarianism 0.8, Authoritarianism 0.2
	wildcard	Martyr
World	player_relation (start)	0.5
	known_events	moses_defection, hop_removal, dysphoria_expansion, romancian_takeover, commonman_formed
	known_figures	Jean Pope Chautlier, Amourie Othella, Krystian Krakk

Moses is the most confrontational NPC in the PNE test data. Maximum assertion (1.0), high self-esteem (0.8), and low cognitive flexibility (0.3) combine to produce a resistance profile that makes authority-based approaches nearly futile. His near-balanced locus of control (0.475) reflects something more considered: outcomes are shaped by both individual action and systemic forces, which grounds his Utilitarian ideology. Decisions serve the Insurgency's collective survival, not personal sentiment.

The Martyr wildcard is the most extreme in the current NPC set. Under ideological pressure it hard-selects Defend Cause Passionately at near-maximum confrontation, temporarily bypassing the standard desire-to-intention flow. It fired twice during the test run.

6.2.3 Troy

Table 6.2.3 — NPC JSON Profile: Troy

Block	**Field**	**Value**
Metadata	Name	Troy
	Age	25
	Faction	Insurgency
	Social Position	Vice
Cognitive	self_esteem	0.2
	locus_of_control	0.85
	cog_flexibility	0.1
Social	assertion	0.8
	conf_indep	0.1
	empathy	0.5
	ideology	*(none)*
	wildcard	*(none)*
World	player_relation (start)	0.5
	known_events	moses_defection
	known_figures	Morisson Moses

Troy is the simplest psychological profile in the test corpus, but perhaps the most difficult to deal with. Very low self-esteem (0.2) makes him sensitive to insults. He has very low cognitive flexibility (0.1), so he is likely to rely on his first impressions of the player. High assertion (0.8) means he fights when under threat. He has no ideology dictionary and no wildcard. His actions are all through the base BDI pipeline with no override. His conf_indep score is 0.1 constituting high conformity; his unwavering fealty to Moses and the Insurgency. Efforts to appeal to the cause work well.

6.3 Cross-NPC Response to the Same Player Input

The most direct evidence that the pipeline responds to NPC personality, rather than producing uniform outputs, comes from comparing how Troy and Morisson Moses processed the identical opening choice: *open_authority* (*"I don't need to justify myself to a door guard. We're fighting the same war. Open the door."*).

Both NPCs received the same player text. Both produced negative emotional valence (−0.45). Beyond that, their pipeline states diverged substantially across every subsequent layer.

Table 6.3.1 — Pipeline State Comparison: Troy vs Morisson Moses, open_authority

Layer	**Troy**	**Morisson Moses**
Bias Type	black_white_thinking	ideological_filter
Internal Thought	"There's no middle ground here. Never was."	"They're framing this wrong. The whole premise is off."
Subjective Belief	"I need to know which side of this they're on before I say another word."	"The way they're framing this tells me they haven't fully reckoned with what this costs."
Desire Type	dominance	protection
Desire Intensity	0.90	0.75
Intention Type	Challenge Back	Defend Cause Passionately
Confrontation Level	0.654	**0.978**
Wildcard Triggered	No	**Yes (Martyr)**
Relation Delta	−0.15	−0.15
NPC Response	*"You picked the wrong side, and you know it. Move."*	*"You've got more questions than answers, pal. Move it or lose your balls."*

Troy's low cognitive flexibility (0.1) and binary worldview produced a *black_white_thinking* bias — the authority challenge was processed as a loyalty binary, generating a dominance desire and a mid-range confrontation level of 0.654. His response was blunt but contained.

Moses processed the same input through an *ideological_filter* bias — his entrenched Utilitarian worldview and maximum assertion caused him to evaluate the challenge not as a loyalty test but as evidence of ideological misalignment. His Martyr wildcard fired, bypassing the normal intention scoring and hardwiring him to *Defend Cause Passionately* at 0.978 confrontation. His response was significantly more explosive.

The divergence is entirely attributable to their personality parameters. The pipeline processed the same input text and produced different bias categories, different desire states, different intention types, radically different confrontation levels, and qualitatively different spoken outputs. Neither response was authored directly; both emerged from the NPCs' respective psychological configurations.

Krakk did not receive open_authority in his test run; he received open_diplomacy instead. The resulting pipeline state is shown in Table B.1 and discussed in 6.4.1 below. The contrast between Krakk's +0.45 positive valence on a diplomatic opener and Troy and Moses's −0.45 negative valence on the authority opener further illustrates how the same scenario produces fundamentally different conversational trajectories depending on both NPC profile and player approach.

6.4 Individual NPC Conversation Logs

6.4.1 Krakk Klikowicz : 2-Turn SUCCEED

Full turn-by-turn BDI breakdown for Krakk Klikowicz: see Appendix B, Table B.1.

Krakk terminated on SUCCEED after two turns, the shortest of three. The diplomatic opening (*open_diplomacy*) prompted *confirmation_bias* as the cognitive filter read his own values into the player's statement: "Their values and mine are close enough to matter". His emotional valence was +0.45, resulting in a desire for information at 0.6 and a Neutral Evaluation intention at confrontation 0.5. His *cog_flexibility* increased by +0.05 after the outcome, as he became slightly more open to what was being said as the discussion went well. The relation delta was +0.10.

The terminal condition for the *concrete_value* choice on turn 2 was reached before a second NPC turn. Final *player_relation*: 0.75. The Inferiority wildcard did not trigger, as the diplomatic tone did not reach the threshold for the authority tone that triggers it, so Krakk's actions remained unmodified by the wildcard. This example illustrates the pipeline's operation for a high-empathy/high-independence NPC using ideologically similar framing.

6.4.2 Morisson Moses: 4-Turn SUCCEED

Full turn-by-turn BDI breakdown for Morisson Moses: see Appendix C, Table C.1.

Moses took four turns to reach SUCCEED (tied for longest). The Martyr wildcard fired on turns 1 and 3, where the choices were framed in ideologically challenging ways (*open_authority* on turn 1, *mutual_challenge* on turn 3). Each wildcard turns chose Defend Cause Passionately at 0.978 confrontation with a passionate outburst, the highest confrontation values in the test corpus.

Turn 2 (*authority_soften*) triggered a shift in the pipeline. The player's confession of being too pushy triggered *ideological_filter* bias again, but this time with positive valence (+0.45), which turned Moses's desire into a lower intensity quest for information (0.55). The wildcard did not trigger this turn; the softening did not reach the ideological pressure threshold. The intention changed to Challenge to Reveal Truth at confrontation 0.9, and the relation improved by +0.08. This was the only personality attribute change in the run of Moses: *cog_flexibility* +0.03, the beginning of a break in his inflexibility.

Turn 3 (*mutual_challenge*) triggered the wildcard again, but the relation was positive, so close to the terminal condition. Turn 4 (*concrete_value*) met it. Moses's final *player_relation* of 0.68 - the lowest SUCCEED score in any run - is the result of opening with an authority challenge on the system's most aggressive NPC, and the cumulative cost. This is an instrumental rather than a collaborative success, reflecive Moses's personality and his parting words: "You had potential with your offers, make good on them or get out of the way."

6.4.3 Troy: 4-Turn SUCCEED

Full turn-by-turn BDI breakdown for Troy: see Appendix D, Table D.1.

Troy's test led to the highest final *player_relation* of the three main tests (1.0) despite starting from the same authority challenge as Moses. But what happened next. Troy's *black_white_thinking* bias on turn 1 generated a dominance desire - he thought the player might be a potential opponent - but the recovery turn (*authority_soften*) generated *confirmation_bias* rather than antagonism. His subjective belief became "They're thinking about this in the same way as me", so the acknowledgement of overreach was interpreted through his loyalty framework as a sign of ideological compatibility, rather than weakness. The desire transitioned to inform, and the relation increased by +0.08.

Turn 3 (*concrete_value*) triggered the same *confirmation_bias* and continued the positive trend, with another +0.15-relation delta. Troy's *cog_flexibility* remained stable across all turns, as per his base value of 0.1 - his lack of flexibility accentuated the new positive assessments. Turn 4 (*mutual_challenge*) was the final turn. His final message - "You have shown initiative; let's see what you can do next." - is one that makes a binary judgement in favour of the player, as is in keeping with his profile.

The difference between Troy's final relation (1.0) and Moses's (0.68) is striking given that both characters ran four turns and began with the same challenge to the leadership. The difference is due to their personality settings: Troy's low independent judgement (*conf_indep* 0.1) made him vulnerable to the shared-cause framing of middle turns whereas Moses (with higher independent judgement and ideological rigidity) was not.

6.5 Comparative Run: Player Build vs. Same NPC

To isolate the effect of player skill distribution on pipeline output, the door_guard_night scenario was run twice against Troy using contrasting *PlayerSkillSet* configurations under matched choice conditions. Build configurations are recorded in Table F.1 (Appendix F).

Run A used an empathy-weighted build (authority: 2, diplomacy: 5, empathy: 9, manipulation: 2). The *open_empathy* opener triggered *ideological_filter* bias with positive valence (+0.45), producing an *information-seeking* desire at intensity 0.65 and a *Neutral Evaluation* intention. Troy's *internal_thought*: *"Finally — someone who understands what we're actually up against."* The *concrete_value* follow-up met the terminal condition. Terminal: **SUCCEED**. Final *player_relation*: 1.0.

Run B used an authority-weighted build (authority: 9, diplomacy: 5, empathy: 2, manipulation: 2). The *open_authority* opener triggered *black_white_thinking* bias with negative valence (-0.45), producing a *dominance* desire at intensity 0.90 and a *Challenge Back* intention at confrontation 0.515. The recovery attempt (*authority_soften*) was processed through *confirmation_bias* but returned a *backpedal_rejected* outcome (*relation_delta*: -0.05). Troy's rigid cognitive model had already categorised the player negatively; the backpedal read as tactical calculation rather than genuine recalibration. Terminal: **FAIL**. Final *player_relation*: 0.75 (An error was made in not clearing the player_relation post Run-A meaning that the relation began above 0.5, however the result is still valuable.)

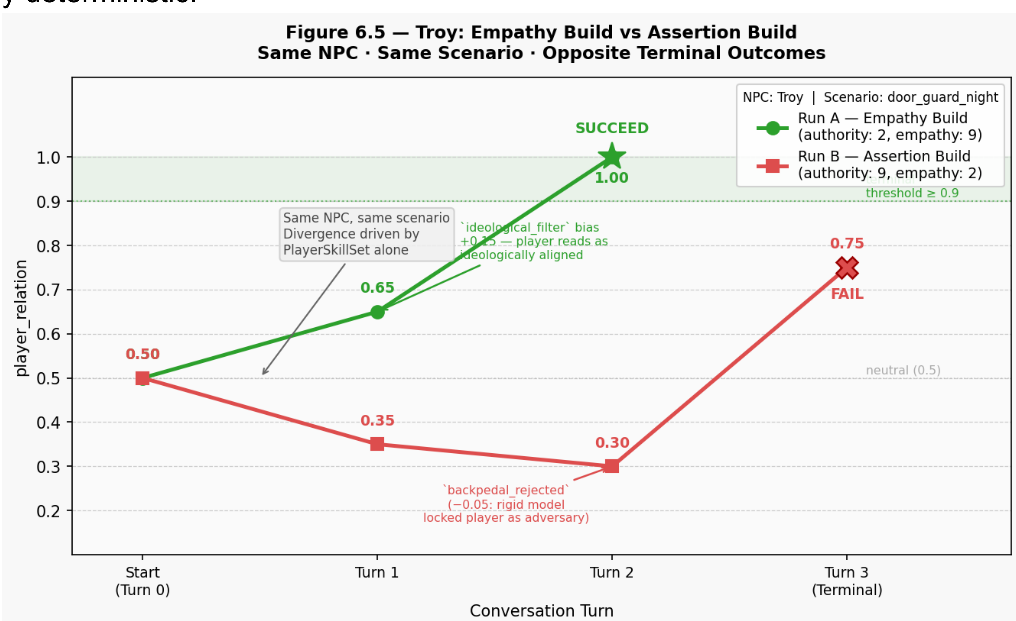
The contrast is clean. Same NPC, same scenario, same choice sequence. Different skill distributions produced different bias activations, different desire states, and opposite terminal outcomes. The pipeline's sensitivity to player build is traceable at every stage.

Table 6.5.1 — Comparative Run Summary: Troy Empathy Build vs Assertion Build

	Run A (Empathy)	**Run B (Assertion)**
Player skill profile	authority: 2, empathy: 9	authority: 9, empathy: 2
Opening choice	*open_empathy*	open_authority
Turn 1 bias type	ideological_filter	black_white_thinking
Turn 1 internal thought	"Finally — someone who understands what we're actually up against."	"There's no middle ground here. Never was."

	Run A (Empathy)	**Run B (Assertion)**
Turn 1 desire type	information-seeking	dominance
Turn 1 desire intensity	0.65	0.90
Turn 1 emotional valence	+0.45	−0.45
Turn 1 relation delta	+0.15	−0.15
Turns to terminal	2	3
Terminal outcome	**SUCCEED**	**FAIL**

The same NPC, the same scenario, and largely the same choice pool produced divergent pipeline states and opposite terminal outcomes. The mechanism is the *PlayerSkillSet*: dice bias determined which choices cleared the skill check, which shaped the input signals reaching the cognitive layer, which cascaded through belief-keyword matching in the desire layer and confrontation scoring in the intention layer, through to the judgement score threshold that triggered terminal routing. The divergence is structural and traceable at every stage. It is not a product of LLM variation , the determining logic, from cognitive through intention layers, is entirely deterministic.



Two-line chart showing Run A (empathy build, green) and Run B (assertion build, red) relation trajectories across turns 0–3. Run A reaches SUCCEED (★) at turn 2 via a clean ideological alignment arc; Run B hits backpedal_rejected at turn 2 and terminates at FAIL (✗) on turn 3. Terminal threshold band and divergence annotation included.

6.6 Terminal Outcome Summary

Table 6.6.1 — Terminal Outcome Summary Across All Test Runs

NPC	Approach	Turns	Wildcard Fires	Final Relation	Terminal
Krakk Klikowicz	Diplomatic	2	0	0.75	SUCCEED
Morisson Moses	Authority → Recovery	4	2 (Martyr)	0.68	SUCCEED
Troy (6.1 run)	Authority → Recovery	4	0	1.0	SUCCEED
Troy (Empathy build)	Empathetic	2	0	1.0	SUCCEED
Troy (Assertion build)	Authority	3	0	0.75	FAIL

Across five test runs, the system produced four **SUCCEED** outcomes and one **FAIL**. The single **FAIL** is the most diagnostically valuable result in the corpus. It demonstrates that terminal routing is not trivially permissive, and that approaching a psychologically resistant NPC carries genuine mechanical consequence. Moses's *player_relation* of 0.68 on his **SUCCEED** run, the lowest in the set, further illustrates that the system distinguishes between clean successes and hard-won ones. The two-turn **SUCCEED** outcomes (Krakk and Troy empathy build) show the other end: an aligned approach against a receptive profile can close a conversation efficiently, without requiring a recovery arc.

The wildcard results are informative in aggregate. Moses's *Martyr* wildcard fired twice in four turns, producing near-maximum confrontation on both occasions. Krakk's *Inferiority* wildcard did not fire at all. Troy carries none. This distribution reflects the design intention: wildcards are personality-extreme overrides that activate under specific conditions, not persistent modifiers that colour every turn. Fired twice in one run, inactive in two others, their behaviour in the corpus is consistent with that role. *Chapter 7 evaluates these results against the system's central architectural claims.*

Chapter 7: Discussion and Analysis

7.1 Validation of Outputs

The central architectural claim of this dissertation is that a BDI pipeline parameterised by NPC personality attributes will produce psychologically consistent internal states, and that those states will be reflected in NPC dialogue output in a verifiable, traceable way. This section evaluates that claim using the test evidence from Chapter 6.

My approach to validation separates two distinct questions that are often conflated in natural language generation systems. The first is whether the pipeline's deterministic layers — cognitive interpretation, desire formation, and intention selection — produce outputs that are internally consistent with the NPC's personality and update in the expected direction across turns. This question can be answered without any reference to the LLM, because those layers do not call it. The second is whether the LLM-generated dialogue reflects the pipeline state it receives. This question is harder to evaluate rigorously, because LLM output is probabilistic. I treat these separately below.

7.1.1 Internal Consistency

Internal consistency, as used here, means that NPC attribute deltas and pipeline state transitions accumulate in the direction the personality model predicts. Not just within a single turn, but across a multi-turn conversation where each turn is processed against an updated state. A pipeline that is internally consistent should produce progressively worsening relation scores and increasingly resistant internal states in response to escalating hostility. Conversely, a recovery sequence following an aggressive opener should produce measurable softening, traceable to specific delta values in the log.

The test data in Chapter 6 supports this directly. The most instructive evidence comes from the Moses run (Table C.1) and Troy's run (Table D.1), both following the same four-turn structure: authority challenge on turn 1, recovery attempt on turn 2, renewed pressure on turn 3, transactional close on turn 4.

In the Moses run, the +0.03 *cog_flexibility* delta on turn 2 is directionally correct. A genuine softening gesture produces a small but non-zero crack in his rigidity. Turn 3's *mutual_challenge* raises his assertion by +0.1 — a high-assertion NPC under ideological pressure gets more confrontational, not less. The magnitudes are modest. The directions are exactly what the profile predicts.

Troy produced the same delta (+0.03 *cog_flexibility* on the recovery turn) despite starting from a base of 0.1. After the shift his flexibility sits at 0.13; Moses's sits at 0.33. Different ceilings, same trigger, both traceable in the log.

Krakk's case reinforces the pattern from the other direction. His +0.05 delta on a constructive opener is larger than the +0.03 recovery deltas seen after hostile openers against Moses and Troy. That difference matters: it reflects dispositional receptivity, not damage repair. The pipeline treated these situations differently, and the numbers show it.

Across all three primary runs, no relation delta or attribute shift moved against what the NPC's profile would predict. That is the consistency claim being made here. Not that the magnitudes are perfectly calibrated as they are not, and that is discussed under limitations, but that the directional behaviour is correct and followable at every turn.

For a prototype, this is the right bar. The pipeline is not producing noise. It is producing structured psychological responses grounded in the parameters each character was given.

Figure 7.1 — NPC Relation Trajectories (see Appendix K). Three-line chart (turn 0–4) showing per-turn player_relation values for each NPC. The chart visualises the divergence between Krakk's diplomatic arc (2-turn SUCCEED), Moses's contested recovery arc (Martyr wildcard annotations at turns 1 and 3, final relation 0.68), and Troy's binary-then-committed arc (final relation 1.0). The terminal threshold band (≥ 0.9) is marked.

7.1.2 Output Coherence

The second validation question is whether LLM-generated dialogue reflects the pipeline state it receives. This is harder to assess. It requires interpreting text, and qualitative interpretation introduces subjectivity that quantitative log data does not. The approach here is to focus on cases where the pipeline state made a specific, strong prediction about the character of the response, and then examine whether the generated text holds up against that prediction. Match or misfire becomes the lens.

The strongest predictions come from the intention layer's *expression* field and the *confrontation_level* value. When a turn produces an *explosive* expression at 0.978 confrontation, the pipeline is asserting near-maximum hostility. That is a specific, falsifiable claim about the text that follows.

The Moses run provides the clearest test cases. Turn 1 (*open_authority*) produced: *Martyr* wildcard triggered, *Defend Cause Passionately* intention, *explosive* expression, confrontation 0.978. Moses's response: *"You've got more questions than answers, pal. Move it or lose your balls."* Not measured — bold, dismissive, shaped by raw reaction rather than deliberation. Consistent with the computed state.

Turn 3 (*mutual_challenge*) fired the wildcard again under identical pipeline conditions. Moses's response: *"That's a start, but you need more than vague promises. Keep it real and tangible from here on out."* More restrained than turn 1, despite the same *explosive* tag and 0.978 confrontation level. The *protection* desire and *ideological_filter* bias are still legible in the text, but the aggression has reduced without explanation. This is a coherence gap , not a failure of the pipeline's logic, but an inconsistency in how forcefully the model enacted its own instructions.

Troy's run demonstrates stronger coherence across all four turns. Turn 1's *black_white_thinking*, *dominance* desire, confrontation 0.654 produced: *"You picked the wrong side, and you know it. Move."* There was no room given. A precise match. Turn 2's shift to *soften_stance*, *information-seeking*, confrontation 0.59 produced: *"That's better. Keep talking."* the model caught the pivot, from threat toward cautious invitation with accuracy. Turn 3's *confirmation_bias*, *information-seeking*, confrontation 0.59 produced: *"That's a start. Keep going but be specific about what you can offer."* Which shows rigidity bending without breaking completely.

What is apparent in Troy's run is not that every response is perfectly calibrated, but that the sequence forms a coherent arc. Hostile, then cautiously open, then conditionally engaged. The tonal trajectory mirrors the relation delta and attribute trajectory logged at the pipeline level. Responses move in step with decisions already recorded internally. That coherence across turns is more telling than isolated precision.

The Moses turn 3 gap needs characterising correctly. It is not evidence of a flawed architecture. The pipeline logic was consistent, the wildcard fired correctly, and the intention was selected through the normal scoring process. The gap exists at the interface between computed output and the model's realisation of that output as text. At qwen2.5:3b scale, intensity signals like *explosive* are sometimes overridden when recent turn history pulls in a different direction. A more capable model would be expected to close this gap without any changes to the pipeline. This is a capability ceiling, not a design failure, and it is consistent with the deployment context analysis in Section 7.2.2.

7.1.3 Terminal Outcome Routing

The third validation question is whether terminal outcomes are reached or missed in a way that is congruent with the player's conversational history, not randomly triggered after a certain number of turns. A pass/fail routing system that operates independently of what the player said would not be player agency, it would just be a timer. The results in Chapter 6 support the argument that this is not the case. Most obvious is the comparative Troy run in Section 6.5. The empathy build (Run A: authority: 2, empathy: 9) reached SUCCEED in two turns. The authority build (Run B: authority: 9, empathy: 2) reached FAIL in three turns. The same NPC, the same conversation, the same choice tree, reach different ends.

The reason is traceable: Run B's authority-biased dice roll gave a negative first impression (*black_white_thinking* bias), which - due to Troy's *cog_flexibility* of 0.1 - reinforced itself, rather than being updated on the backpedal. The judgement score did not reach threshold; the conversation ended at **FAIL** with a -0.05-relation delta for the *backpedal_rejected* choice.

The empathy-weighted dice in Run A produced an *open_empathy* opener. The cognitive layer interpreted this as confirming a shared ideology, rather than threatening, which resulted in an *ideological_filter* bias, +0.45 positive valence, information-seeking desire, and a +0.15-relation delta. Troy's thought bubble - "Finally - someone who understands what we're really up against." - reflects the belief model with the player's tone confirming Troy's model, as would be expected with his *confirmation_bias* bias under positive input. The second option (*concrete_value*) was terminal without a recovery arc. **SUCCEED** in two turns.

The point of this example is to demonstrate that the terminal routing is based on judgement score, which is based on dice outcomes, which are based on skill distribution and choices. The routing isn't random, nor is it a turn counter. A player who matches their choices to an NPC's psychological profile will terminate conversations quickly; a player who misreads the profile will run out of turns or do damage that prevents success. The links are visible in the logs.

The Moses run adds a further dimension. Moses succeeded with a final *player_relation* of 0.68 - the lowest **SUCCEED** in the corpus. He was successful because the relation trajectory crossed a terminal threshold at turn 4, but barely. His final message, "You have the potential to make good offers, but now put it to the test or get out of the way", is more transactional than cooperative. The terminal routing made a distinction between this ungracious success and Troy's 1.0 final relation, even though both ended with the **SUCCEED** flag. The system doesn't just report success; it reports the quality of the success in the final relation value that captures the conversational history in a form suitable for downstream narrative routing or world-state systems.

To me, the single **FAIL** in the test data is uniquely important. If the data were all **SUCCEEDs**, then this would demonstrate that the pipeline can only end conversations, but not that it can "work" either direction. The **FAIL** shows that it is fully functional in both directions. The terminal conversation responds to the player's conversational history and if the player's "strategy" does not match the personality of the NPC, they will not reach the threshold, no matter how many turns are left.

(See Figure 6.5 — Appendix L — for the dual-build relation trajectory chart, which makes the divergence between Run A and Run B visually legible across turns.)

7.2 Justification of Achievements

7.2.1 The System as a Functional Prototype

With the engine, I didn’t really plan for an entire dialogue engine for a commercial title; I didn’t have a clear initial goal outside of experimentation. However, I was certain that I wanted to demonstrate that a BDI cognitive architecture alongside psychological constructs could use LLM text generation for psychologically grounded NPC dialogue for narratives. By that standard, I consider the project a success.

What I delivered is a functional prototype of an incredibly ambitious system, one that processes player input through four psychologically grounded pipeline stages, produces a fully articulated internal NPC state on every turn, constrains a locally hosted LLM to give voice to that state, and routes narrative outcomes through a judgement-score finite state machine.

I built this alone, within an academic project timeline, while simultaneously learning Ollama, BDI modelling and learning to implement a REST and WebSocket API. I ensured that the system functions coherently end-to-end and it will reach terminal outcomes. This I am incredibly proud of.

The more honest measure of the project's achievement is not whether it is production ready, It’s conceptual validity. For me, I believe it to be worth continued exploration. I believe the PNE has genuine value beyond its academic context and would warrant further development as a standalone middleware product. The architecture is highly plausible, and the core pipeline logic is fully separable from any specific game engine. These design choices exist because I always had a real use case in mind.

7.2.2 Where the System Works

The clearest way I can explore what the system achieves is to be specific about where it works and where it does not.

The PNE is best suited to games where dialogue is light. In contexts where conversation contributes personality and texture without carrying the weight of a complex narrative. *Stardew Valley* (ConcernedApe, 2016) is the example I keep returning to. That game's dialogue is deliberately simple: villagers respond to the player's gift choices and seasonal events with short, characterful exchanges that display personality and relationship state without too much narrative consequence.

The gameplay loop is agricultural and social simulation; the narrative is rather complementary. An engine like the PNE would transform those exchanges substantially. It gives villagers a structured internal model of their relationship with the player, producing responses that reflect accumulated social history, and allowing the same player action to land differently depending on the NPC's psychological profile. The dialogue quality ceiling of qwen2.5:3b is not a disqualifying limitation in that context; short, characterful exchanges are precisely what smaller quantised models handle most reliably.

The system is not suited, in its current form, to games where dialogue is critical. This is exemplified by narratives such as *Fallout 4* (Bethesda Game Studios, 2015). The companion characters in that title require nuanced memory, tonal precision across emotionally loaded scenes, and consistent personality expression across dozens of hours of play. The qwen2.5:3b model cannot reliably sustain this quality level, and the PNE's current scenario graph architecture does not model long-term narrative continuity at the required depth. I do not consider this a failure of the architecture's design principles. It is a scope boundary defined by hardware constraints and a single-developer timeline, and it is one I am transparent about.

The distinction between these two contexts is, for me, the most useful thing my evaluation can establish. The question is not about "does it work?" but moreso were.

7.3 Research Question Revisited

The central question I posed in Chapter 1 was: *to what extent can a BDI cognitive architecture, instantiated over a structured NPC personality model and constrained LLM text generation, produce NPC dialogue that is simultaneously psychologically grounded, narratively coherent, and genuinely responsive to player agency?*

My answer is *yes,* but with careful considerations.

The pipeline provides psychologically plausible NPC behaviour. The cognition layer converts the player choice to a belief, which is influenced by an NPC personality; the desire layer converts the belief to a desire and the intention layer chooses from a small set of behaviours, customised by the author. An NPC will never respond to the player with a string hard-coded into the game (unless Ollama fails), or a pattern-matching response. NPC responses are the result of the final belief, which changes as a function of the NPC personality and player choice. This is the major architectural claim of my dissertation, and it is clear from the logs that this is the case.

Choice is not entirely removed. The judgement score is an amalgamation of evidence over several turns; the terminal conditions are based on the state of the conversation and the two dice for skill check is the random element in the game but not at the cost of choice.

A player who makes choices that are consistent with the profile of a given NPC is not the same as a player who makes wrong inferences; this is evident between the steps in the logs. The limiting factor is narrative consistency, and it is not a design choice but = The LLM's hardware. Respecting is not 100% at the level of the qwen2.5:3b. Sometimes the response is not in the right tone or is commentary. At times, even not honouring the inner state nuance. When this happens, the pipeline's rich context is lost in the speech.

It's important to note this is not a pipeline failure, but a model constraint. This is remedied by boosting the model with more energy on small hardware and some engine tweaks.

7.4 Critique

7.4.1 The Logging Gap

The biggest methodological weakness of my project is that the amount of testing I did is far more than the amount of test evidence I stored. I tested the system through a vast number of conversational scenarios over the course of the project, resulting in an ongoing flow of pipeline data which informed every significant redesign recorded in Chapter 5.
Most of this was viewed in the terminal and lost. The files that constitute the main documentary evidence in Chapters 4 and 6 are only a tiny subset of the testing I've done, there were at minimum 100 tests ran.

At the time, this felt acceptable. I was more concerned with experimenting. I wanted to identify architectural inefficiencies, not log the results. It was a problem when I needed to show variations in the NPCs' abilities for the evaluation phase. Scenarios that I have tested many times and found to be effective couldn't be demonstrated in retrospect because I did not log the results as consistently.

The reality is that my narrative evaluation is based on less evidence than my experience would indicate it should be. If I did this project again the first change I would make would be to ensure that there was an automatic logging protocol in place from the very beginning of the prototyping process, currently it is optional per scenario. The ideal change would be that all tests should be serialised and saved with a structured note of what was being tested and the observation. This would have turned my validation into a strong section of the dissertation.

7.4.2 The Wildcard System

The wildcard system is probably the most notable example of a feature that was overrun by the project. The original behaviour of the wildcard was to modify the prompt used to call Ollama: a straightforward way to pass the LLM direct instructions for the wildcard in question. Amourie Othella's *Napoleon* wildcard, for example, would change the NPC's response to be a dominant one if the player's tone of authority exceeded a certain threshold, passing the tone change as part of the prompt to Ollama.

The *CognitiveThoughtMatcher* changed all that. Prior to the design change, every turn made two calls to Ollama: cognitive thought generation, and NPC dialogue. The wildcard was designed to insert itself into the latter. The *CognitiveThoughtMatcher* combined the two —

replacing the LLM-based cognitive layer with deterministic pattern matching and consolidating all LLM interaction into a single dialogue generation call. This eliminated the mechanism the wildcard relied on.
The *wildcard_triggered* flag is still present in the pipeline, triggers specific intention templates in the intention registry, and is sent to the LLM as part of the *BehaviouralIntention* output. But without a new prompt augmentation layer for each wildcard type, the model does not reliably produce the desired behavioural change.

Redesigning the wildcard for the new single-call architecture would have required time to develop a new use case, time that unavailable. Future development would focus on redesigning the wildcard's prompt interaction layer, likely as an extension of the *min_response*/*max_response* anchor mechanism from the outcome layer.

The following chapter draws these threads together in a conclusion and reflection.

Chapter 8: Conclusion and Future Work

8.1 Conclusion

The PNE delivers what branching dialogue trees structurally cannot: a narrative space that is not fully pre-authored. Each NPC response is the product of an internal state that varies continuously with personality and conversational history — not a fixed string retrieved from a decision graph. Whether a character grants access, shifts alliance, or closes off entirely is a function of who they are and what they have interpreted across the conversation, not of which node a graph route to.

The most significant architectural decision was decoupling cognitive reasoning from language generation. This was not planned. When the LLM handled both cognitive thought and surface dialogue, hallucination and latency affected two layers simultaneously. The *CognitiveThoughtMatcher* resolved both at once: reasoning became reliable and instantaneous, and the LLM's single remaining role became more constrained and therefore more consistent.

The separation of what the NPC thinks from how the NPC speaks is now the central architectural success of this work. This was reached through iterative problem-solving, not top-down design. The CHANGELOG entries trace every revision from a specific observed failure to a specific architectural response, making the development process itself part of the evidence.

8.2 Originality of Contributions

The PNE is not the first system to apply BDI architecture to NPC behaviour, nor the first to use a language model for dialogue generation. Its originality lies in the specific combination of properties it assembles, and in the practical feasibility conditions it satisfies that prior work does not.

Mateas and Stern's *Façade* (2003) demonstrated drama-managed interactive narrative but required years of development, cloud-adjacent infrastructure, and produced failures when player input exceeded its NLU capacity. Park et al.'s Generative Agents (2023) demonstrated LLM-driven social simulation at scale, but with a cloud-hosted model doing both reasoning and generation, a coupling that makes the quality of social behaviour entirely dependent on model capability, and the deployment incompatible with offline consumer games. Traditional BDI implementations in the Bordini, Hübner and Wooldridge (2007) tradition address goal-directed agents in structured task environments, not open conversational dialogue with continuous personality parameters.

No prior work combines a grounding in cognitive and social psychology literature; deterministic, author-defined BDI reasoning; locally deployable LLM text generation; a closed intention vocabulary that makes FSM routing reliable; and a deployment architecture that is engine-agnostic and accessible to solo and small-team developers.

The emergent narrative property that results from this combination is also novel in its framing. Prior emergent narrative systems produce emergence as a side-effect of autonomous character behaviour (Aylett, 1999) or player-side freedom (Murray, 1997). The PNE locates emergence on the NPC side, within a structured but non-scripted internal model: the same player action can produce a trust-building trajectory with one NPC and a terminal failure with another, not because the developer wrote both outcomes, but because the pipeline computed them from different psychological starting points. This is a new point in the design space.

8.3 Future Work

Three directions for continued development are clear from the current implementation, ordered by practical impact.

Dedicated Authoring Interface

The most immediately limiting constraint for a developer integrating the PNE is the requirement to write and edit NPC profiles, scenarios, and world data in raw JSON. The HTML character creator (*character_creator*.html) represents the beginning of a solution, but it is partial as it does not cover scenario graph authoring, thought-template management, or world event configuration. A full desktop authoring application, comparable in scope to Qt Designer, where graphical interfaces over structured data remove the need to understand file syntax. This lowers the barrier to entry for narrative designers and solo developers. This is the single most impactful investment for making the PNE practically usable beyond its current developer audience.

Cloud Infrastructure Variant

The current architecture is deliberately local-first: Ollama runs on the player's machine, no external requests are made, and offline deployment is guaranteed. This is the right trade-off for the target use case, but it is not the only valid one.

A cloud-hosted variant of the pipeline would replace the Ollama call with an API to a larger model. Potentially delivering better dialogue quality and removing the VRAM constraint for players on lower-spec hardware, on the condition that the game requires an internet connection. The deterministic pipeline layers would remain unchanged; only the text generation endpoint would differ. Architecturally this is straightforward; the commercial and privacy trade-offs it introduces are a design decision for any developer integrating the system.

Completing the Wildcard System

As documented in Section 7.4.2, the wildcard architecture is structurally present but not fully realised. The *wildcard_triggered* boolean reaches the LLM prompt, but without per-wildcard generation constraints the model does not reliably alter its output in response. A wildcard-specific prompt augmentation layer — analogous to the *min_response*/*max_response* anchor mechanism in the outcome layer — would complete the feature and unlock the full range of NPC archetypes the system is designed to support.

8.4 Reflection

Before beginning this project, I had no meaningful understanding of BDI models. I had not heard of it whatsoever, the idea that the philosophical framework Bratman (1987) developed for practical reasoning could be instantiated as a Python pipeline was not a connection I had made. Working through the implementation from the ground up changed that. I now have a genuine interest in cognitive computing that I did not have before: the question of how understanding cognitive structures can be replicated inside computation to create systems that behave in psychologically interpretable ways feels like a field worth working in, and this project is the reason I now hold that passion.

My biggest technical surprise was Ollama. I had no knowledge of it before the project began, and my initial assumption was that integrating a local language model would be the most technically demanding component of the system. It was arguably the easiest.

The difficulty was not the integration; it was the selection. Understanding my hardware constraints (4 GB VRAM), researching which models fit within them at 4-bit quantisation, and evaluating the quality ceiling of *qwen2.5:3b* against the requirements of conversational dialogue generation required genuine research and reflection. The lesson was not that the technology was hard, but that the judgement around it, knowing which tool is right for which constraint, is what is.

What I would tell a second-year student considering a project like this is that the passion comes first. The PNE exists because I had a real use case I cared about from the beginning; not a problem assigned to me, but a system I wanted to exist. That motivation is what sustained the iterative cycle of running several tests, researching cognitive psychology, and iterating multiple versions through failure. It is also what made the difficulty feel like an invitation rather than an obstacle. This is the most satisfying work I have done since entering university. I am proud of what I built, I understand its limitations clearly, and I am looking forward to continuing it beyond the academic context in which it began.

References

Aylett, R. (1999) 'Narrative in virtual environments — towards emergent narrative', *AAAI Fall Symposium on Narrative Intelligence*, Technical Report FS-99-01. Menlo Park: AAAI Press, pp. 83–86.

Beck, A.T. (1979) *Cognitive therapy of depression*. New York: Guilford Press.

Bethesda Game Studios (2015) *Fallout 4* [Video game]. Rockville, MD: Bethesda Softworks.

Bordini, R.H., Hübner, J.F. and Wooldridge, M. (2007) *Programming multi-agent systems in AgentSpeak using Jason*. Chichester: Wiley.

Bratman, M. (1987) *Intention, plans, and practical reason*. Cambridge, MA: Harvard University Press.

Cialdini, R.B. (1984) *Influence: the psychology of persuasion*. New York: Harper Collins.

ConcernedApe (2016) *Stardew Valley* [Video game]. Self-published.

Davis, M.H. (1983) 'Measuring individual differences in empathy: evidence for a multidimensional approach', *Journal of Personality and Social Psychology*, 44(1), pp. 113–126.

Ellis, A. (1962) *Reason and emotion in psychotherapy*. New York: Lyle Stuart.

Festinger, L. (1954) 'A theory of social comparison processes', *Human Relations*, 7(2), pp. 117–140.

Georgeff, M., Pell, B., Pollack, M., Tambe, M. and Wooldridge, M. (1999) 'The belief-desire-intention model of agency', in Müller, J.P., Rao, A.S. and Singh, M.P. (eds.) *Intelligent agents V: agents theories, architectures, and languages*. Lecture Notes in Computer Science, vol. 1555. Berlin: Springer, pp. 1–10.

Goffman, E. (1959) *The presentation of self in everyday life*. New York: Anchor Books.

Guimond, S. (ed.) (2006) *Social comparison and social psychology: understanding cognition, intergroup relations and culture*. Cambridge: Cambridge University Press.

Heider, F. (1958) *The psychology of interpersonal relations*. New York: Wiley.

Mateas, M. and Stern, A. (2003) 'Façade: an experiment in building a fully-realized interactive drama', *Proceedings of the Game Developers Conference*, San Jose, CA.

Murray, J.H. (1997) *Hamlet on the holodeck: the future of narrative in cyberspace*. New York: Free Press.

Neisser, U. (1967) *Cognitive psychology*. New York: Appleton-Century-Crofts.

Park, J.S., O'Brien, J.C., Cai, C.J., Morris, M.R., Liang, P. and Bernstein, M.S. (2023) 'Generative agents: interactive simulacra of human behavior', *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST '23)*. New York: ACM.

Qwen Team, Alibaba Cloud (2024) *Qwen2.5: a party of foundation models*. Technical report. Available at: https://ollama.com/library/qwen2.5:3b (Accessed: March 2026).

Rao, A.S. and Georgeff, M.P. (1995) 'BDI agents: from theory to practice', *Proceedings of the First International Conference on Multi-Agent Systems (ICMAS-95)*, San Francisco, CA. Menlo Park: AAAI Press, pp. 312–319.

Telltale Games (2012) *The Walking Dead: Season One* [Video game]. San Rafael, CA: Telltale Games.

Wooldridge, M. (2000) *Reasoning about rational agents*. Cambridge, MA: MIT Press.

ZA/UM (2019) *Disco Elysium* [Video game]. London: ZA/UM.

Appendices

Appendix A — PNE Pipeline Stage Overview

Referenced in Section 5.1.1.

Table A.1 — PNE Pipeline Stages: Purpose and Output

Each turn of dialogue passes through seven sequential stages. The table below describes what each stage does and what it produces.

Stage	Name	What it does	What it produces
I	NPC Intent Layer	Loads the NPC's identity, backstory, long-term goals, and the possible outcomes for the scene. Sets the rules everything else must follow.	The NPC's internal goal state; the available terminal endings; player skill levels and difficulty settings.
II	Choice Selection	The player picks a dialogue option. The system reads the tone of that choice (diplomatic, commanding, emotional, etc.) and packages it for the pipeline.	A structured player input with the choice text, communication style, and tone scores.
IIa	Skill Check	Two biased dice are rolled — one for the player, one for the NPC. The player wins on a tie. Player skill and NPC personality	A pass/fail result. On pass: the positive outcome set is used. On fail: the negative set. The result is also fed into the AI prompt later.

Stage	Name	What it does	What it produces
		both influence the dice weights.	
III	Cognitive Interpretation	The NPC privately interprets the player's words through its own psychological lens — confidence, rigidity, empathy all shape what it "hears". This is never spoken aloud.	An internal thought, a subjective belief, an emotional reaction, and a cognitive bias label.
IV	Desire Formation	The NPC's internal belief is translated into a goal. Seven patterns map belief content to desire types (e.g. wanting information, seeking connection, or feeling threatened). The NPC's cognitive bias can sharpen or redirect the desire.	A desire type (information-seeking / affiliation / protection / dominance) and a goal description.
V	Socialisation Filter	The desire type is matched against a fixed list of named NPC behaviours (the Intention Registry). The NPC's social personality — assertiveness, empathy, faction — determines which behaviour fits best. Wildcards (e.g. Napoleon, Martyr) can override this entirely.	A named behavioural intention (e.g. "Acknowledge with Reservation"), a confrontation level, and an emotional tone for delivery.

Stage	Name	What it does	What it produces
VI	Conversational Output	The intention selects an outcome from the scenario. Relation and attitude scores are updated. The AI model (Ollama) is then given the NPC's full profile, the BDI result, the scene direction, and the dice outcome — and generates one in-character spoken line.	The NPC's dialogue response; updated relation and stance values; a momentum tag used to filter the next set of player choices.
VII	Terminal Check	After each turn, the engine checks whether an ending condition has been met — based on accumulated judgement score, relation level, turn count, or what the player has said. If yes, a final AI line is generated and the scene closes. If no, the player is shown new choices and the loop continues from Stage II.	Either a terminal outcome (the scene ends with a concrete world result) or a route to the next dialogue node.

Appendix B — BDI Conversation Log: Krystian 'Krakk' Klikowicz

Referenced in Sections 6.4.1 and 6.3. Source file: tests/Dissertation Tests/krakk_klikowicz_6.1.json.

Table B.1 — Conversation Log: Krystian 'Krakk' Klikowicz (krakk_klikowicz_6.1.json)

Scenario: door_guard_night | **Turns:** 2 | **Terminal:** SUCCEED

NPC Profile

Section	Field	Value	Notes
Metadata	Name	Krystian 'Krakk' Klikowicz	Runner faction boss; enforces decentralised coordination.
	Age	28	—
Cognitive	Self-Esteem	0.5	Moderate — neither easily dominated nor overconfident.
	Locus of Control	0.8	High (internal) — sees himself as the agent of outcomes; authority appeals carry some weight.
	Cognitive Flexibility	0.4	Low-moderate — resistant to rapid ideological shifts, but not closed off.
Social	Assertion	0.3	Low — probes rather than confronts.
	Conf / Independence	0.9	Very high — acts on personal judgement almost exclusively.
	Empathy	0.8	High — responsive to emotional sincerity and personal appeals.

Section	Field	Value	Notes
Ideology	Libertarianism	0.9	Strong — autonomy and freedom are primary motivators.
	Individualism	0.1	Weak — not self-serving; operates for collective Runner benefit.
Wildcard	—	Inferiority	Deep sense of needing to prove worth; can hard-override intention flow under pressure.
Faction	—	Runners	Boss-level social position.

Turn-by-Turn BDI Breakdown

Turn 1

Layer	Field	Value
Player Input	Choice	open_diplomacy
	Text	*"I understand your caution — you'd be a fool not to have it. Let me explain how what I bring serves what you're trying to protect."*
Cognitive	Internal Thought	"They're thinking the same way I do. Can't fake that."
	Subjective Belief	"Their values and mine are close enough to matter."
	Self-Esteem	0.5

Layer	Field	Value
	Locus of Control	0.8
	Cog Flexibility	0.457
	Emotional Valence	**+0.45** (positive)
	Bias Type	confirmation_bias
Desire	Immediate Desire	"Evaluate if they align with our mission"
	Desire Type	information-seeking
	Intensity	0.6
Intention	Type	Neutral Evaluation
	Confrontation Level	0.5
	Expression	analytical
	Wildcard Triggered	No
Outcome	ID	propose_alliance
	Stance Delta	cog_flexibility +0.05
	Relation Delta	+0.10
	Intention Shift	Propose Alliance
NPC Response	—	*"You make your point clear, but I still sense a hidden agenda. Let's see how far you can go without crossing the line."*

Turn 2

Layer	Field	Value
Player Input	Choice	concrete_value

Layer	Field	Value
	Text	*"I can bring you things you can't get here — food, medicine, weapons. That's not words, that's logistics. What do you need?"*
NPC Response	—	*(Terminal outcome triggered — no further NPC turn logged)*

Terminal Outcome

Field	Value
Result	SUCCEED
World Effect	Guard grants entry to the Insurgency hideout. Trust established.
Final Dialogue	*"I see your value, but my defenses won't always be so high. Allow me to show you what I mean with this first transaction."*
Final player_relation	0.75

Appendix C — BDI Conversation Log: Morisson Moses

Referenced in Sections 6.4.2 and 6.3. Source file: tests/Dissertation Tests/morisson_moses_6.1.json.

Table C.1 — Conversation Log: Morisson Moses (morisson_moses_6.1.json)

Scenario: door_guard_night | **Turns:** 4 | **Terminal:** SUCCEED

NPC Profile

Section	Field	Value	Notes
Metadata	Name	Morisson Moses	Insurgency leader; former Romancian military officer.
	Age	35	—
Cognitive	Self-Esteem	0.8	High — confident and resistant to manipulation or flattery.
	Locus of Control	0.475	Near-balanced; leans external — systemic forces shape outcomes as much as individuals.
	Cognitive Flexibility	0.3	Low — ideologically entrenched; hard to shift without concrete proof.
Social	Assertion	1.0 (final)	Maximum — confrontational, direct, unyielding under pressure.
	Conf / Independence	0.7	Leans independent — guided by principle over consensus.
	Empathy	0.45	Below average — pragmatic rather than emotionally responsive.
Ideology	Utilitarianism	0.8	Strong — decisions must serve the greater good of the Insurgency.

Section	Field	Value	Notes
	Authoritarianism	0.2	Weak — accepts command structure but not blind hierarchy.
Wildcard	—	Martyr	Willing to sacrifice himself or others for the cause; can dominate intention selection under ideological pressure.
Faction	—	Insurgency	Boss-level social position.

Turn-by-Turn BDI Breakdown

Turn 1

Layer	Field	Value
Player Input	Choice	open_authority
	Text	*"I don't need to justify myself to a door guard. We're fighting the same war. Open the door."*
Cognitive	Internal Thought	"They're framing this wrong. The whole premise is off."
	Subjective Belief	"The way they're framing this tells me they haven't fully reckoned with what this costs."
	Self-Esteem	0.8
	Locus of Control	0.475

Layer	Field	Value
	Cog Flexibility	0.3
	Emotional Valence	**−0.45** (negative)
	Bias Type	ideological_filter
Desire	Immediate Desire	"Maintain distance and protect boundaries"
	Desire Type	protection
	Intensity	0.75
Intention	Type	Defend Cause Passionately
	Confrontation Level	0.978
	Expression	explosive
	Wildcard Triggered	**Yes**
Outcome	ID	reject_authority
	Stance Delta	*(none)*
	Relation Delta	−0.15
	Intention Shift	Reject Authority
NPC Response	—	*"You've got more questions than answers, pal. Move it or lose your balls."*

Turn 2

Layer	Field	Value
Player Input	Choice	authority_soften
	Text	*"I pushed too hard. Look — I need in. What do you actually need from me?"*

Layer	Field	Value
Cognitive	Internal Thought	"I need to know if they actually believe this or if they just know what to say."
	Subjective Belief	"The pitch is solid. The real question is whether they'd still be here when it got harder."
	Emotional Valence	**+0.45** (positive)
	Bias Type	ideological_filter
Desire	Immediate Desire	"Acknowledge but remain guarded"
	Desire Type	information-seeking
	Intensity	0.55
Intention	Type	Challenge to Reveal Truth
	Confrontation Level	0.9
	Expression	direct
	Wildcard Triggered	No
Outcome	ID	soften_stance
	Stance Delta	cog_flexibility +0.03
	Relation Delta	+0.08
	Intention Shift	Soften Stance
NPC Response	—	*"That's a start, but we'll see how far that goes. Keep your proposals concrete and specific."*

Turn 3

Layer	Field	Value
Player Input	Choice	mutual_challenge
	Text	*"You've been testing me since I walked up. Fine. But you and I are fighting the same war. Either let me prove that, or we both lose."*
Cognitive	Internal Thought	"They're framing this wrong. The whole premise is off."
	Subjective Belief	"Their framing is ideologically misaligned — they're not seeing what I see."
	Emotional Valence	**+0.45** (positive)
	Bias Type	ideological_filter
Desire	Immediate Desire	"Maintain distance and protect boundaries"
	Desire Type	protection
	Intensity	0.75
Intention	Type	Defend Cause Passionately
	Confrontation Level	0.978
	Expression	explosive
	Wildcard Triggered	**Yes**
Outcome	ID	challenge_accepted
	Stance Delta	assertion +0.1
	Relation Delta	+0.10
	Intention Shift	Challenge to Reveal Truth
NPC Response	—	*"That's a start, but you need more than vague promises.*

Layer	Field	Value
		Keep it real and tangible from here on out."

Turn 4

Layer	Field	Value
Player Input	Choice	concrete_value
	Text	*"I can bring you things you can't get here — food, medicine, weapons. That's not words, that's logistics. What do you need?"*
NPC Response	—	*(Terminal outcome triggered — no further NPC turn logged)*

Terminal Outcome

Field	Value
Result	SUCCEED
World Effect	Guard grants entry to the Insurgency hideout. Trust established.
Final Dialogue	*"You showed potential with your offers, now deliver on them or step back."*
Final player_relation	0.68

Appendix D — BDI Conversation Log: Troy

Referenced in Sections 6.4.3 and 6.3. Source file: tests/Dissertation Tests/troy_6.1.json.

Table D.1 — Conversation Log: Troy (troy_6.1.json)

Scenario: door_guard_night | **Turns:** 4 | **Terminal:** SUCCEED

NPC Profile

Section	Field	Value	Notes
Metadata	Name	Troy	One of the earliest Insurgency members; devoutly loyal to Moses.
	Age	25	—
Cognitive	Self-Esteem	0.2	Very low — highly reactive to perceived disrespect or dominance.
	Locus of Control	0.85	High (internal) — believes outcomes are determined by action; authority is credible to him.
	Cognitive Flexibility	0.1 (base)	Very low — binary worldview; slow to shift from initial read on a person.
Social	Assertion	0.8	High — will confront directly when he perceives threat.
	Conf / Independence	0.1	Very conformist — loyalty to Moses overrides independent judgement.
	Empathy	0.5	Moderate — can respond to sincerity

Section	Field	Value	Notes
			but does not lead with it.
Ideology	—	*(none defined)*	No explicit ideological framework; motivation is personal loyalty.
Wildcard	—	*(none)*	No wildcard modifier — intentions follow normal BDI flow.
Faction	—	Insurgency	Vice-level social position (second tier).

Turn-by-Turn BDI Breakdown

Turn 1

Layer	Field	Value
Player Input	Choice	open_authority
	Text	*"I don't need to justify myself to a door guard. We're fighting the same war. Open the door."*
Cognitive	Internal Thought	"There's no middle ground here. Never was."
	Subjective Belief	"I need to know which side of this they're on before I say another word."
	Self-Esteem	0.2
	Locus of Control	0.85
	Cog Flexibility	0.1

Layer	Field	Value
	Emotional Valence	**−0.45** (negative)
	Bias Type	black_white_thinking
Desire	Immediate Desire	"Maintain distance and protect boundaries"
	Desire Type	dominance
	Intensity	0.9
Intention	Type	Challenge Back
	Confrontation Level	0.654
	Expression	assertive
	Wildcard Triggered	No
Outcome	ID	reject_authority
	Stance Delta	*(none)*
	Relation Delta	−0.15
	Intention Shift	Reject Authority
NPC Response	—	*"You picked the wrong side, and you know it. Move."*

Turn 2

Layer	Field	Value
Player Input	Choice	authority_soften
	Text	*"I pushed too hard. Look — I need in. What do you actually need from me?"*
Cognitive	Internal Thought	"Either they believe it or they've done their homework. Either works."

Layer	Field	Value
	Subjective Belief	"They're framing this through the same lens I use."
	Emotional Valence	**+0.45** (positive)
	Bias Type	confirmation_bias
Desire	Immediate Desire	"Maintain distance and protect boundaries"
	Desire Type	information-seeking
	Intensity	0.7
Intention	Type	Neutral Response
	Confrontation Level	0.59
	Expression	direct
	Wildcard Triggered	No
Outcome	ID	soften_stance
	Stance Delta	cog_flexibility +0.03
	Relation Delta	+0.08
	Intention Shift	Soften Stance
NPC Response	—	*"That's better. Keep talking."*

Turn 3

Layer	Field	Value
Player Input	Choice	concrete_value
	Text	*"I can bring you things you can't get here — food, medicine, weapons. That's not words, that's logistics. What do you need?"*

Layer	Field	Value
Cognitive	Internal Thought	"Either they believe it or they've done their homework. Either works."
	Subjective Belief	"They're framing this through the same lens I use."
	Emotional Valence	**+0.45** (positive)
	Bias Type	confirmation_bias
Desire	Immediate Desire	"Maintain distance and protect boundaries"
	Desire Type	information-seeking
	Intensity	0.7
Intention	Type	Neutral Response
	Confrontation Level	0.59
	Expression	direct
	Wildcard Triggered	No
Outcome	ID	transactional_deal
	Stance Delta	*(none)*
	Relation Delta	+0.15
	Intention Shift	Transactional Agreement
NPC Response	—	*"That's a start. Keep going, but be specific about what you can offer."*

Turn 4

Layer	Field	Value
Player Input	Choice	mutual_challenge

Layer	Field	Value
	Text	*"You've been testing me since I walked up. Fine. But you and I are fighting the same war. Either let me prove that, or we both lose."*
NPC Response	—	*(Terminal outcome triggered — no further NPC turn logged)*

Terminal Outcome

Field	Value
Result	SUCCEED
World Effect	Guard grants entry to the Insurgency hideout. Trust established.
Final Dialogue	*"You have shown initiative; let's see what you can do next."*
Final player_relation	1.0

Appendix E — Development Timeline: Major System Milestones

Referenced in Section 5.3.1 (Table E.1).

Table E.1 — Development Timeline: Major System Milestones

Date	Milestone	What was introduced	Why it mattered
16 Nov 2025	Initial project created	Repository initialised. First commit established the project structure.	Starting point — nothing functional yet.
16 Nov 2025	Core data models (PNE_Models.py v1)	CognitivePersonalityModel (self-esteem, locus of control, cognitive flexibility), SocialPersonalityModel (assertion, empathy, ideology), and WorldPerceptionModel (history, relation, known events) defined. NPCModel wraps all three.	Defined the psychological architecture that every subsequent layer was built on top of. These classes remain the backbone of the system today.
17 Nov 2025	Pipeline started + Ollama connected	pipeline.py scaffolded. qwen2.5:3b running locally via Ollama wired in to generate NPC dialogue. First small-scale tests ran successfully.	First time the system produced an actual NPC response. Before this, everything was data models with no behaviour.
22 Nov 2025	Outcome model added	An Outcome type system introduced to represent the consequences of a player choice — attribute changes, relation shifts, and narrative transitions. A data-parsing bug	Without outcomes, choices had no mechanical effect on NPC state. This gave player input a concrete, persistent impact for the first time.

Date	Milestone	What was introduced	Why it mattered
		was found and fixed the same day.	
5 Feb 2026	Package restructure	The single narrative_engine.py file split into a proper Python package: engine.py (orchestrator), session.py (conversation state), cli.py (terminal interface), scenario_loader.py, transition_resolver.py, choice_filter.py, dialogue_coherence.py. Old prototype files moved to depricated/.	Critical architectural step. The monolithic file could not support multi-NPC sessions, an API layer, or independently testable components. This restructure made all later development possible.
5 Feb 2026	Scenario JSON restructured	Scenario nodes updated to explicitly reference which NPC is active per node, rather than the engine assuming a single NPC throughout.	Enabled genuine multi-NPC scenarios and made the dialogue tree engine NPC-agnostic — a scenario file no longer needs to know anything specific about the NPC it runs against.
2 Mar 2026	Intention Registry introduced	A closed vocabulary of named NPC behavioural intentions defined (e.g. "Carefully Question Motives", "Propose Alliance", "Establish Boundaries"). The socialisation layer	Made BDI pipeline output predictable and author-controllable. Scenario designers could write transition conditions that reliably trigger on specific intention names, rather than hoping the LLM

Date	Milestone	What was introduced	Why it mattered
		now scores player–NPC interaction against these templates rather than generating free-form intention text.	produced matching text.
2 Mar 2026	Dialogue coherence + choice filtering	DialogueMomentumFilter and ChoiceFilter introduced. Remaining player choices are scored on conversational coherence across four dimensions — momentum alignment (40%), stage appropriateness (30%), anti-repetition (20%), and relation plausibility (10%) — before being presented to the player. Choices scoring below 0.3 are suppressed.	Prevented the player from ignoring what the NPC just said. Contextually incoherent choices are hidden, making conversation feel like a real exchange rather than a static menu.
2 Mar 2026	World context added to Ollama prompts	NPCs received a snapshot of world history (factions, known events, known figures) injected into the Ollama background prompt. Previously they only had their own personal history and the conversation so far.	Grounded NPC dialogue in the world's political and social context. An NPC aware of the Dysphoria, the Commonman, and the Azraelean War responds substantively differently than one with no world knowledge.

Date	Milestone	What was introduced	Why it mattered
3 Mar 2026 (morning)	Pre-FSM checkpoint	Stable snapshot taken before the FSM rewrite. Scenario JSON restructured so each node directly references active NPCs. Several multi-turn test conversations recorded (Amourie Othella, Morisson Moses).	Preserved a working baseline before a major rewrite. The recorded test conversations directly informed the FSM transition design.
3 Mar 2026 (afternoon)	FSM routing + scene direction	Scenario nodes gained proper transition conditions: turn count, relation thresholds, and intention-keyword gates. The engine picks the first matching transition; unmatched turns stay on the current node. Ollama now receives the node's npc_dialogue_prompt as authorial scene direction alongside BDI state. Recovery mode added — a failed dice roll queues a second-chance set of choices before a path is permanently locked.	Replaced a naive turn counter with a proper stateful conversation engine. For the first time, what the player said — not just how many turns had passed — determined where the story routed.
3 Mar 2026 (evening)	Judgement tracker + new entry point	A judgement score (0–100) introduced per NPC. Each dice outcome shifts it up	Made risk genuinely meaningful. A player who consistently attempts difficult

Date	Milestone	What was introduced	Why it mattered
		or down, with a risk multiplier applied when the pre-roll success probability was below 50% — making bold, low-odds choices carry amplified consequences. judgement replaced player_relation as the primary FSM routing driver. NE.py created as the clean top-level entry point; the old monolithic narrative_engine.py removed.	choices feels the cumulative weight of that pattern in where the story routes. Relation still exists but is now a secondary, complementary signal.
4 Mar 2026	2-dice skill check system	Player and NPC each roll one biased d6 — player wins on a tie. Bias derived from player skill level and NPC resistance threshold (calculated from NPC psychological attributes). success_pct displayed analytically before each choice. Difficulty presets added: SIMPLE (+15% player bias), STANDARD (0%), STRICT (−15%). Failed-choice pruning introduced — paths that fail both the main roll and the recovery	Replaced a binary pass/fail with a probabilistic system grounded in NPC psychology. A highly assertive NPC is genuinely harder to command; a highly empathetic NPC is genuinely easier to reach emotionally. Player skill and NPC personality directly determine the odds.

Date	Milestone	What was introduced	Why it mattered
		roll are permanently removed from future turns.	
9 Mar 2026	Cognitive Thought Matcher	CognitiveThoughtMatcher introduced with 810 scored thought templates in cognitive_thoughts.json. Instead of calling Ollama for every cognitive interpretation, the class scores each template against the player's tone signals (authority, diplomacy, empathy, manipulation) and the NPC's psychological attributes, selecting the highest-scoring match above a 0.35 threshold. Falls back to cynical_realism if nothing qualifies. cognitive.py and desire.py substantially rewritten to consume the matcher's output.	Removed the LLM from the cognitive interpretation layer entirely. Interpretation became deterministic, fast, and testable. Two NPCs with different psychological profiles now provably produce different internal thoughts in response to the same player input — verifiable without running the LLM.
9 Mar 2026	REST + WebSocket API layer	A full api/ package introduced: FastAPI server (main.py), in-memory session store (session_store.py), Pydantic request/response schemas	Decoupled the engine from the command line entirely. Any game engine — Unity, Godot, Unreal — can drive a conversation session over HTTP and WebSocket

Date	Milestone	What was introduced	Why it mattered
		(schema.py), WebSocket turn handler (ws_handler.py), and NPC state updater (npc_updater.py). Ollama token streaming bridged through WebSockets so NPC dialogue arrives word-by-word in real time.	without touching Python or understanding the BDI internals.
9 Mar 2026	Unity + Godot client libraries	Three C# files (PNEClient.cs, PNEDialogueUI.cs, PNETypes.cs) and a full Unity integration guide added under docs/unity/. A Godot 4 GDScript client (api_client_godot.gd) added alongside. Install scripts (install.bat, install.sh) added to the repo root.	Provided game developers with drop-in integration layers so the PNE could be embedded in a real project without writing API client code from scratch. The install scripts reduced setup to a single command on both Windows and Unix.

Appendix F — Player Build Configurations: Troy Comparative Run

Referenced in Section 6.5 (Table F.1).

Table F.1 — Player skill build configurations used in the Troy comparative run (Section 6.5)

	Run A — Empathy Build	Run B — Assertion Build
Authority	2	9
Diplomacy	5	5
Empathy	9	2
Manipulation	2	2
Opening choice	open_empathy	open_authority
Terminal outcome	SUCCEED	FAIL
Turns to terminal	2	3
Final player_relation	1.0	0.75

Run A targeted Troy's susceptibility to appeals framed around shared cause and ideological alignment. Run B led with maximum authority pressure against an NPC whose low cognitive flexibility (0.1) meant an initial negative categorisation reinforced itself rather than updating on recovery. The divergent outcomes demonstrate that player skill distribution, not choice text alone, determines terminal routing.

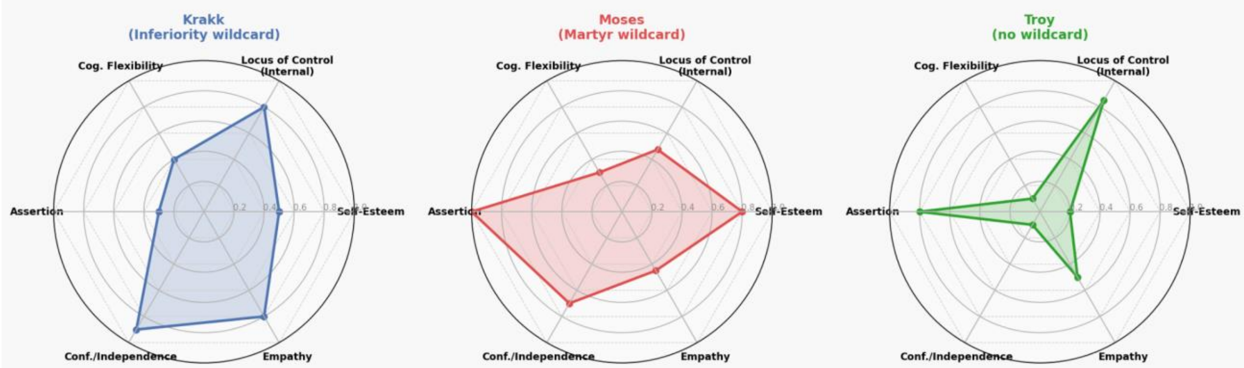
Appendix G — NPC Personality Radar Charts

Referenced in Section 5.1.4 (Figure 6.2). Image file in submitted code repository: Dissertation/Literature Material/Drafts/Graphs/npc_radar_charts.png.

Radar charts plotting the six core personality dimensions for Krakk, Moses, and Troy on a normalised [0, 1] scale. Generated from live NPC JSON profiles using generate_radar_charts.py (same Graphs directory).

Figure 6.2 — NPC Personality Profiles: Krakk, Moses, Troy

Figure 6.2 — NPC Personality Profiles: Krakk, Moses, Troy



NPC	Self-Esteem	LoC (Internal)	Cog. Flex.	Assertion	Conf./Indep.	Empathy	Wildcard
Krakk	0.50	0.80	0.40	0.30	0.90	0.80	Inferiority
Moses	0.80	0.475	0.30	1.00	0.70	0.45	Martyr
Troy	0.20	0.85	0.10	0.80	0.10	0.50	*(none)*

Key contrasts: Krakk's high empathy and independence against low assertion (receptive but autonomous); Moses's maximum assertion and strong self-esteem against near-zero cognitive flexibility (unyielding ideologue); Troy's low self-esteem paired with high assertion and near-zero independence (loyal aggressor, easily destabilised by authority challenges).

Appendix H — Pipeline Architecture: Prototype vs Current System

Referenced in Section 5.3.1 (Table 5.3.1b).

Full comparison table including per-metric summary: see Dissertation/Literature Material/Drafts/Graphs/5.3.1 Pipeline Architecture Comparison.md (Note to marker: this file is included in the submitted code repository.).

Appendix I — Prior Systems Comparison

Referenced in Section 3.7 (Table 3.7) and Section 8.2.

Property	PNE (this work)	Façade (Mateas & Stern, 2003)	Generative Agents (Park et al., 2023)	Jason BDI (Bordini et al., 2007)
Architecture	BDI pipeline (Belief → Desire → Intention) + LLM text generation	Drama-managed NLU + behaviour tree	LLM-driven memory + reflection + planning	AgentSpeak interpreter; formal BDI goal stack
Cognitive grounding	Ellis A-B-C, Heider attribution, Beck distortions — instantiated as 810 parameterised templates	Discourse Acts and beat structure; no psychological personality model	No explicit personality model; behaviour emerges from LLM free-form generation	Logical beliefs and goals; no psychological grounding
NPC personality model	Three-layer continuous parameter space (cognitive, social, world); 6 numeric dimensions + wildcard	None — character behaviour driven by authored beat graph	Seeded persona description in natural language	Logical belief base; no continuous personality dimensions
Text generation	Local LLM (Ollama `qwen2.5:3b`); constrained by closed intention vocabulary and min/max anchors	Rule-based NLG from authored templates	Cloud-hosted GPT-4 generates all agent outputs	None — AgentSpeak is an action-selection language, not a dialogue generator
Reasoning layer	Deterministic (`CognitiveThoughtMatcher`); author-controllable; sub-millisecond	Deterministic drama manager; required years of authoring effort	LLM performs reasoning and generation in a single call — no separation	Deterministic AgentSpeak interpreter; goal selection is formal and auditable
Narrative control	FSM with judgement-score routing; intention registry is a design document; full author override	High — drama manager controls story arc explicitly	Low — emergent from LLM; no guaranteed	High for task environments; not designed for open conversational narrative

			narrative structure	
Offline deployment	Yes — Ollama runs locally; no network dependency	Requires PC-native executable; circa-2003 scale infrastructure	No — requires cloud API (GPT-4); unsuitable for offline consumer games	Yes — Java-based runtime; no cloud dependency
Target use case	Real-time game NPC dialogue; mid-to-high end consumer hardware	Interactive drama (research prototype); not commercially integrated	Social simulation research; not designed for real-time game integration	Multi-agent task environments; not designed for NPC dialogue
Engine integration	Engine-agnostic REST + WebSocket API; tested with Python client	Standalone application	Standalone simulation	Standalone AgentSpeak runtime
Developer accessibility	JSON NPC profiles; HTML character creator; no pipeline code required for integration	Very high authoring cost (years of development per experience)	Requires cloud API key + prompt engineering	Requires AgentSpeak language knowledge; formal verification background useful
Emergent narrative locus	NPC side — personality parameters determine trajectory; same player input diverges across NPC profiles	Drama-managed story arc — emergence controlled by author via beat graph	Agent side — emergent social behaviour from LLM interaction	Not applicable — task-environment agents, not narrative
Published evaluation	Structured log analysis + internal state inspection (Chapters 6–7)	User studies; qualitative player experience analysis	Turing-test-style human evaluation of social plausibility	Formal verification; task-completion benchmarks

Appendix J — Emotional Valence Computation Rules

Referenced in Section 4.2.2 (Table 4.2Referenced in Section 4.2.2 (Table 4.2.2).

Full table with worked example: see Dissertation/Literature Material/Drafts/Graphs/4.2.2 Emotional Valence Rules Table.md (Note to marker: this file is included in the submitted code repository.).

The six rules encode the personality-behaviour mappings from Ellis (1962), Heider (1958), Guimond (2006), and Davis (1983) as conditional additive adjustments to a starting valence of 0.0. Rules are non-exclusive — multiple may fire on the same turn. The result is clamped to [−1.0, 1.0] before being passed to the desire layer.

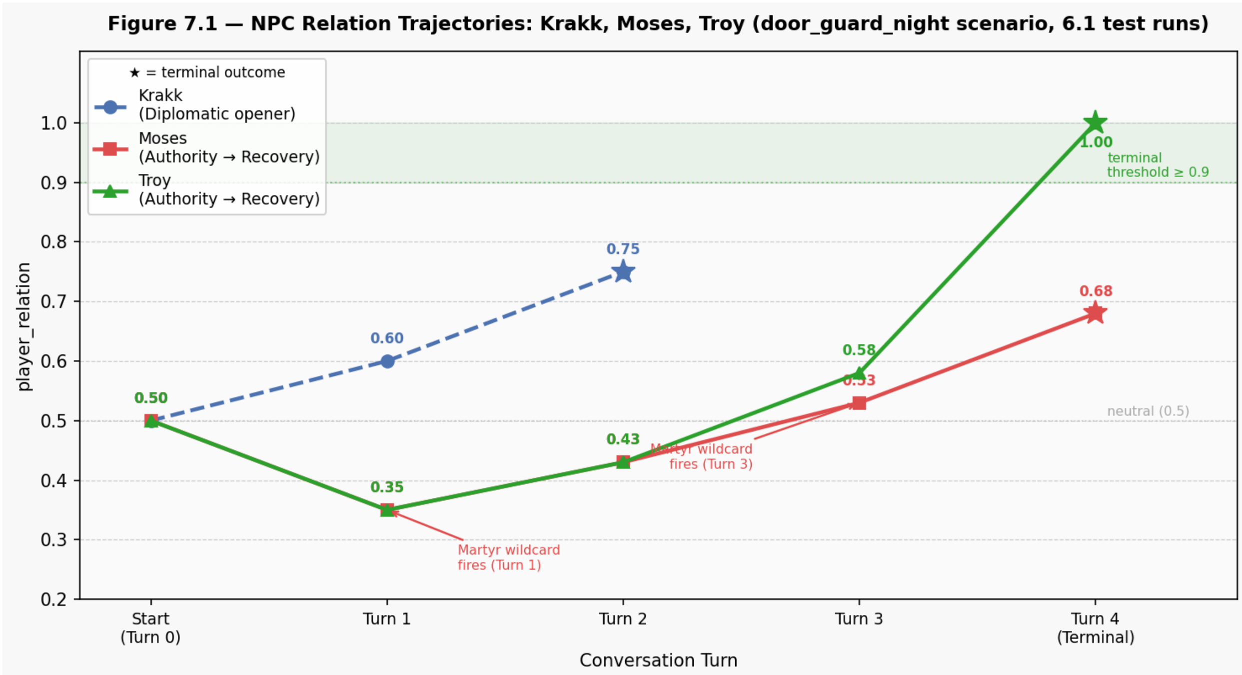
NPC Condition	**Tone**	**Formula**	**Grounding**
self_esteem < 0.4	Authority	− tone × 0.3	Ellis (1962)
self_esteem < 0.4	Manipulation	− tone × 0.5	Ellis (1962)
locus_of_control < 0.5	Authority	− tone × 0.4	Heider (1958)
cog_flexibility > 0.6	Diplomacy	+ tone × 0.4	Guimond (2006)
cog_flexibility > 0.6	Empathy	+ tone × 0.3	Davis (1983)
cog_flexibility < 0.4	Diplomacy	− tone × 0.2	Ellis (1962)

Appendix K — NPC Relation Trajectory Chart

Referenced in Section 7.1.1 (Figure 7.1). Image file in submitted code repository: Dissertation/Literature Material/Drafts/Graphs/npc_trajectory_chart.png.

Line chart generated from tests/Dissertation Tests/ routing logs using generate_trajectory_chart.py (same Graphs directory). Plots player_relation at each turn for Krakk (diplomatic opener, 2 turns), Moses (authority opener, 4 turns, Martyr wildcard fires twice), and Troy (authority opener, 4 turns).

Figure 7.1 — NPC Relation Trajectories



Turn	Krakk	Moses	Troy
Start	0.50	0.50	0.50
1	0.60	0.35	0.35
2	**0.75 X**	0.43	0.43
3	—	0.53	0.58
4 (Terminal)	—	**0.68 X**	**1.00 X**

X = terminal SUCCEED. Krakk reaches terminal on turn 2 via diplomatic approach; Moses and Troy both open with authority challenges but diverge across the recovery arc due to differing personality parameters.

Appendix L — Troy Dual-Build Comparison Chart

Referenced in Section 6.5 (Figure 6.5) and Section 7.1.3.

Figure 6.5 — Troy: Empathy Build vs Assertion Build

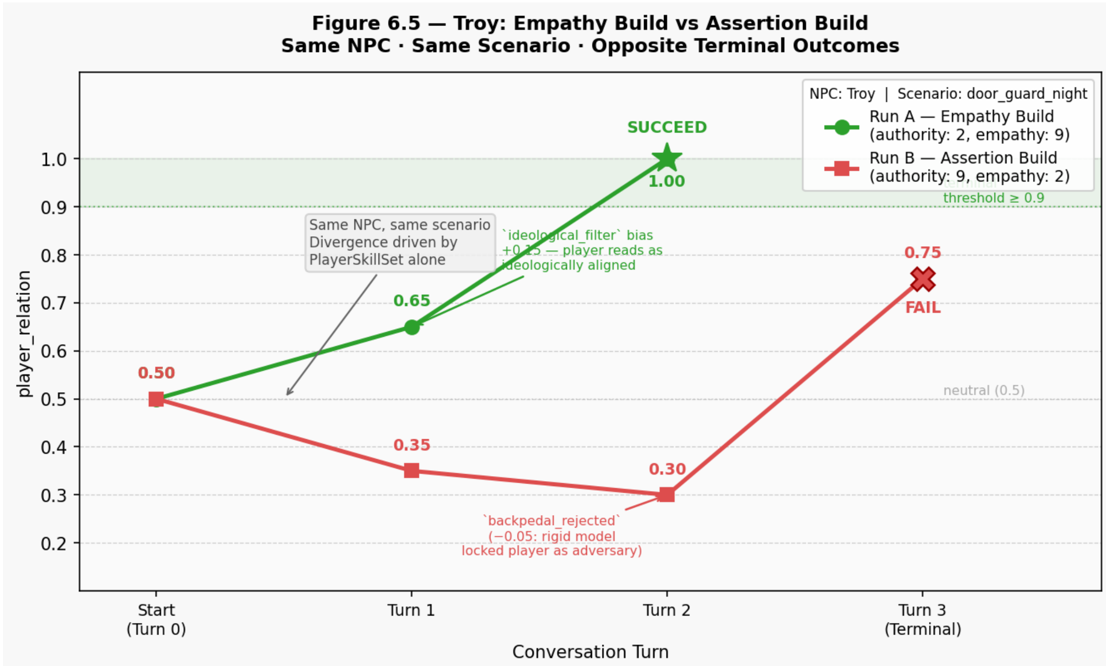


Figure 6.5 — Troy: Empathy Build vs Assertion Build
Same NPC · Same Scenario · Opposite Terminal Outcomes

Turn	Run A — Empathy Build	Run B — Assertion Build
Start	0.50	0.50
1	0.65 (+0.15 — ideological_filter, positive valence)	0.35 (−0.15 — black_white_thinking, negative valence)
2	**1.00 ★ SUCCEED**	0.30 (−0.05 — backpedal_rejected)
3	—	**0.75 ✗ FAIL**

Generated from generate_dual_build_chart.py (Graphs directory). Source data from §6.5 / Table 6.5.1.